Deploying Active Directory Domain Controller in a Virtual Lab Environment

Prepared by: Katelyn S
Date: May 2026

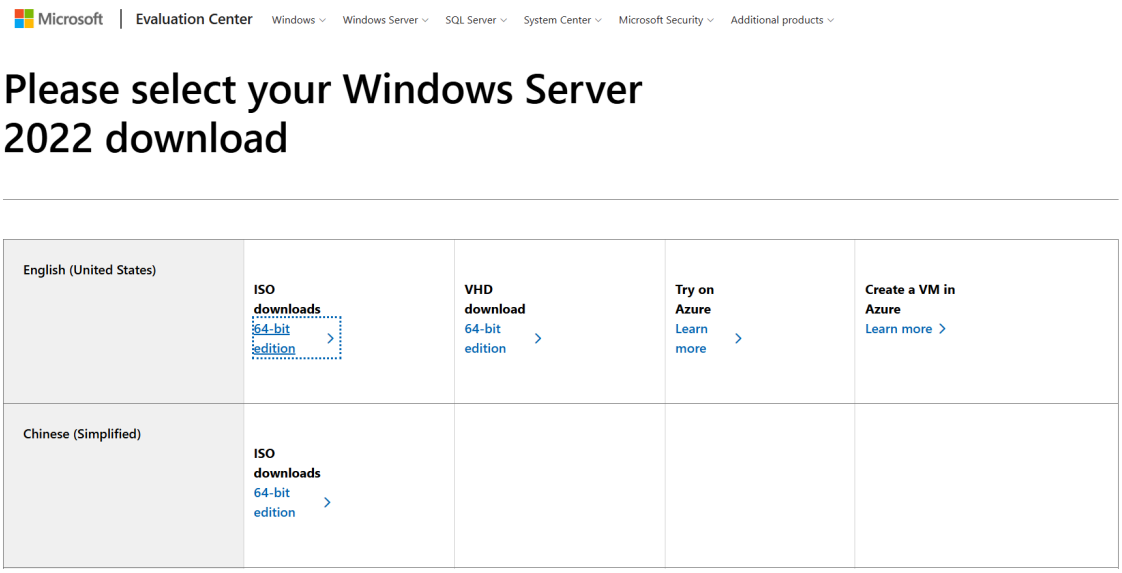
Objective: To deploy Active Directory domain controller in a virtual lab environment using VirtualBox, Windows Server 2022, and Windows 11 Pro. By deploying Active Directory domain control, it simulates organizational structure with users and organizational units which is then validated on Windows 11 client.

Tools used: VirtualBox, Windows Server 2022 , Windows 11 Pro

Steps: Preinstalled VirtualBox and Windows 11 > Installed and Configured Domain Controller on Windows Server 2022 > Added OU and Users > Tested User Log In on Windows 11 Pro to verify the client machine was joined to the domain

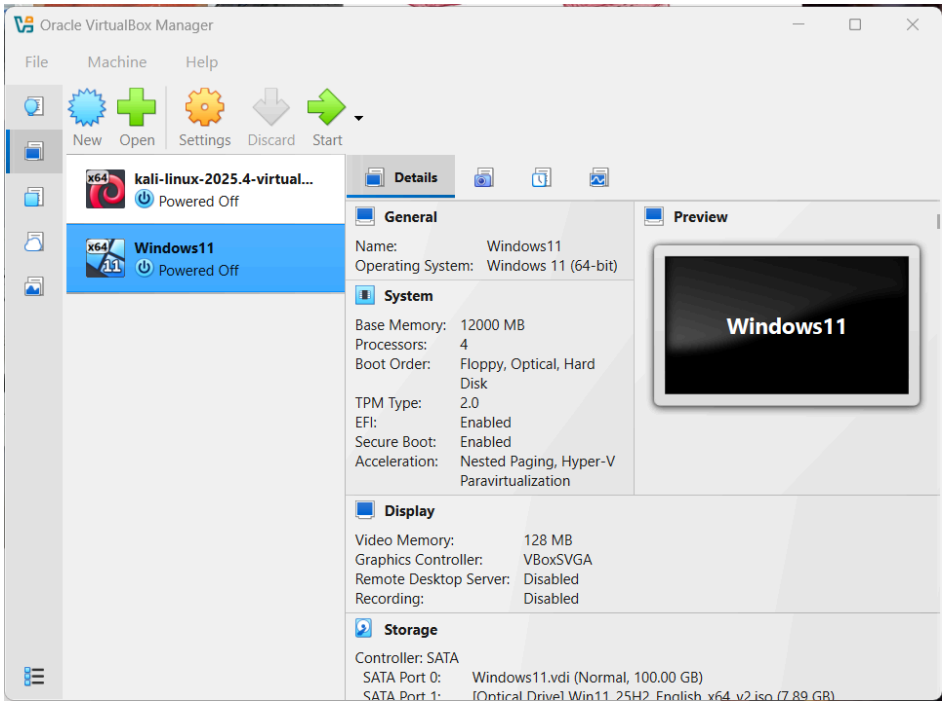
Summary: The lab was successful in deploying an Active Directory domain controller on Windows Server 2022. This allowed me to add organizational units and users to the domain. I was also able to verify Windows 11 Pro connected properly with Windows Server 2022 and recognized the domain.

First, I went to my search engine and typed in windows server 2022 iso and clicked on the link for microsoft. I clicked new to the english row under ISO “64-bit edition”

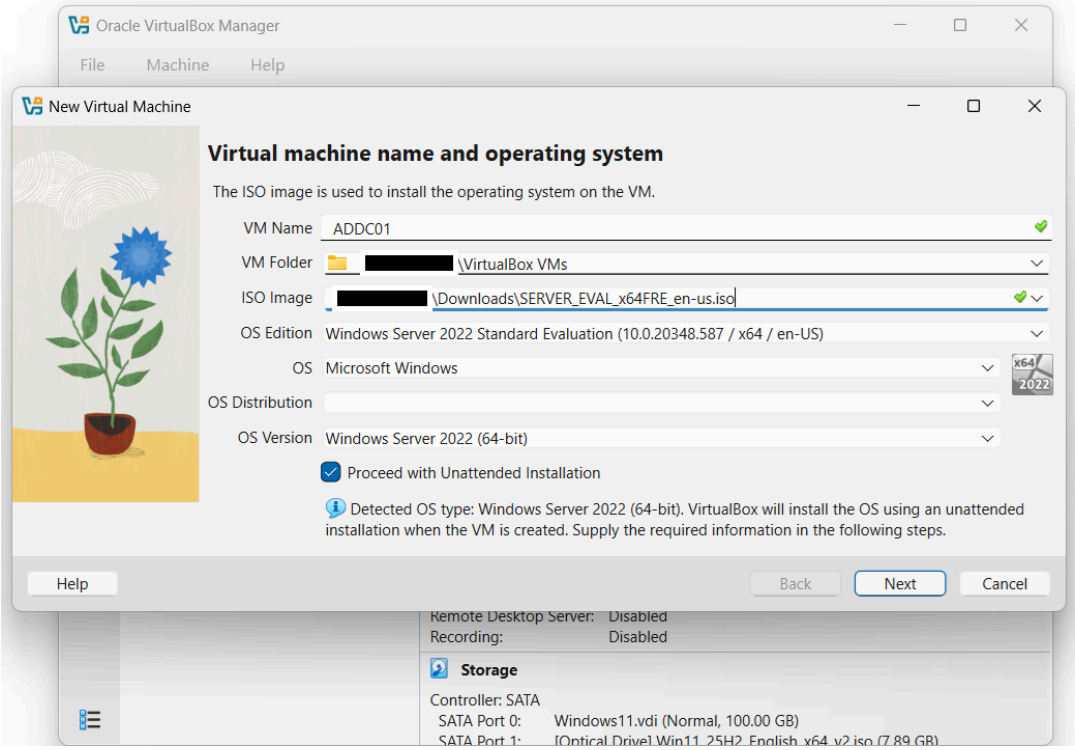


The hyperlink will take you to file explorer where you can save the iso file and start the download process.

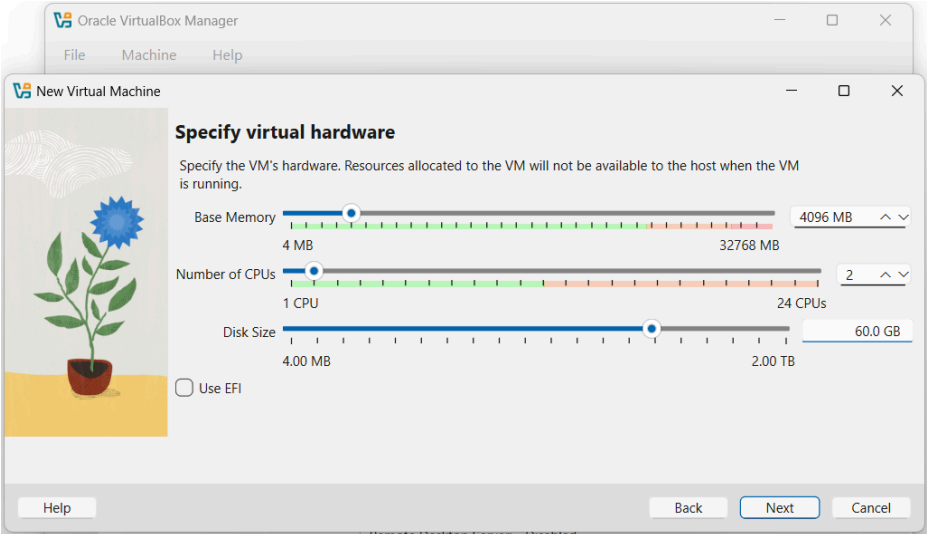
Once finished I went to virtual box and clicked on new



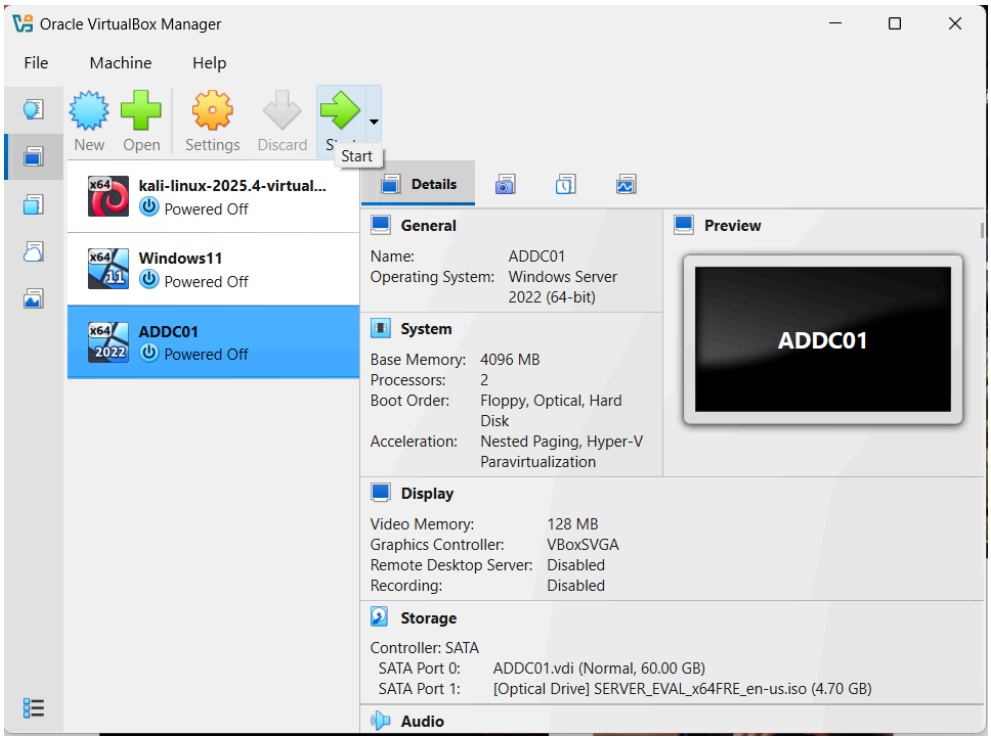
Created the VM Name and located the ISO file for ISO Image



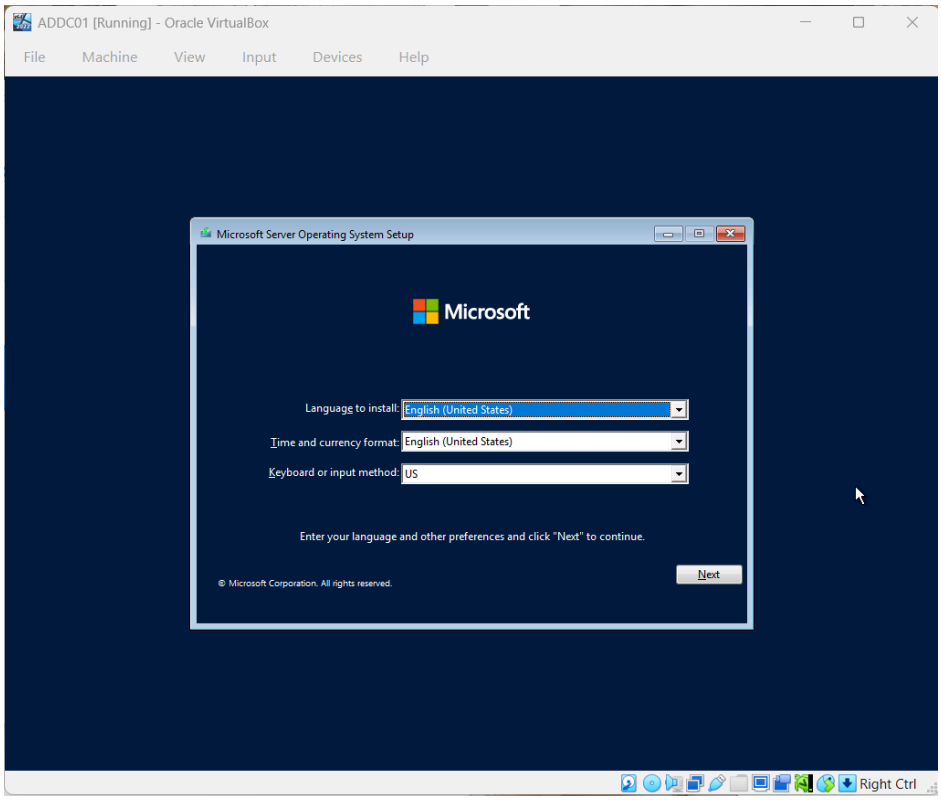
Unchecked the *Proceed with Unattended Installation* box and hit **Next**
I, then, customized the base memory, # of CPUs and Disk Size to my specific needs and hit next



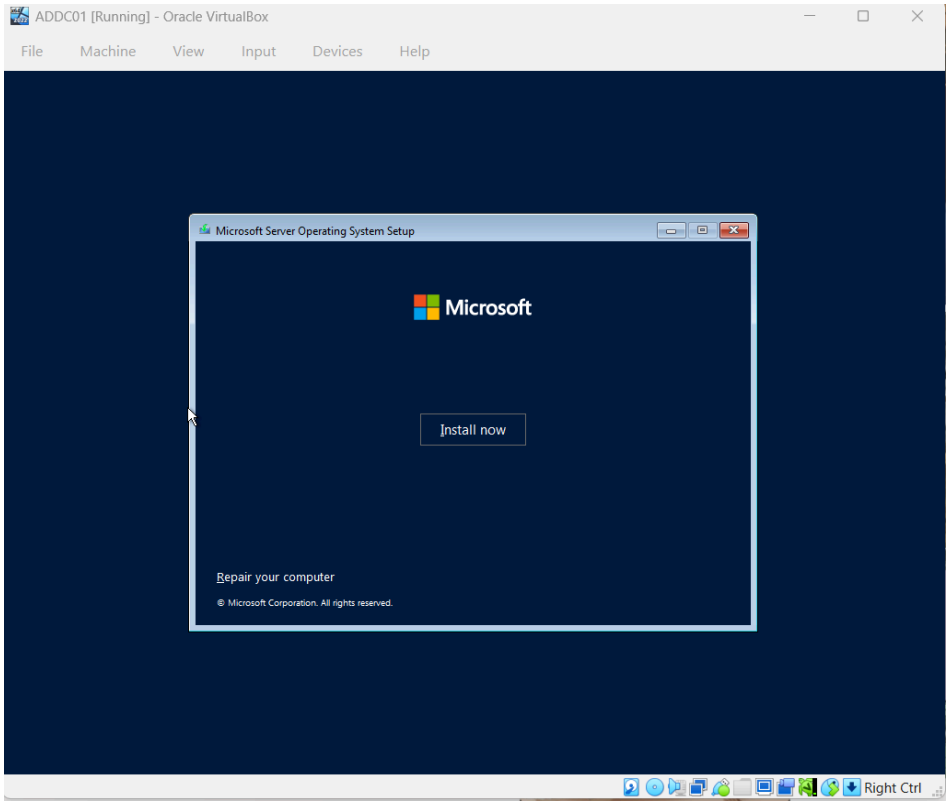
Verified that all the information is correct and hit **Finish**
Hit Start to start up the Windows Server (ADDC01)



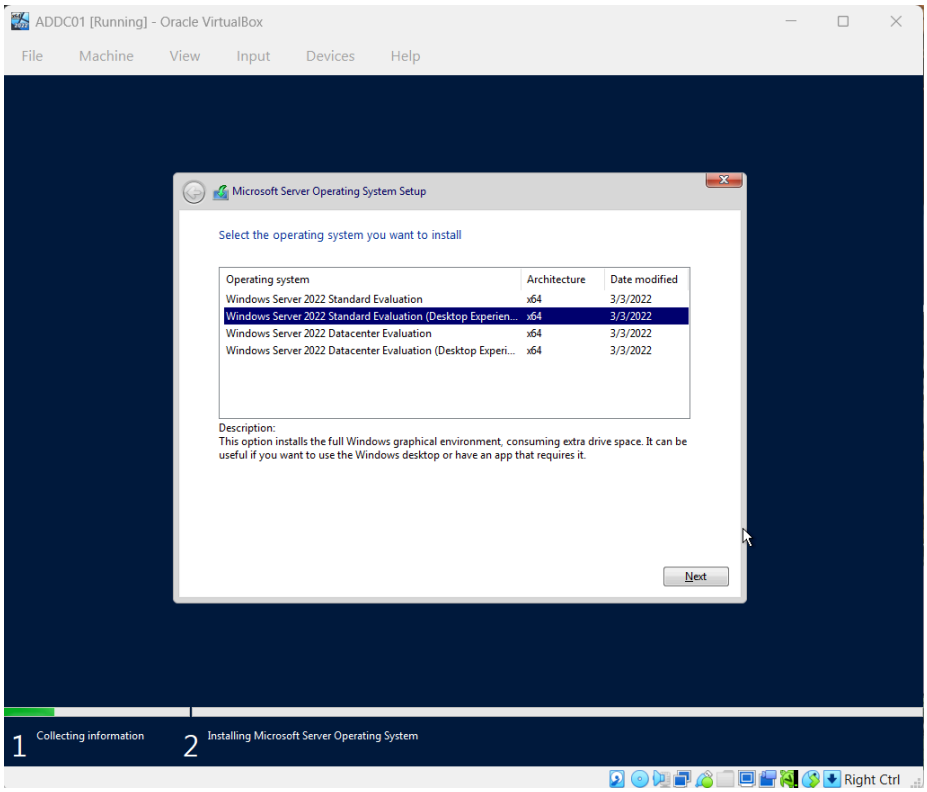
Hit **Next**



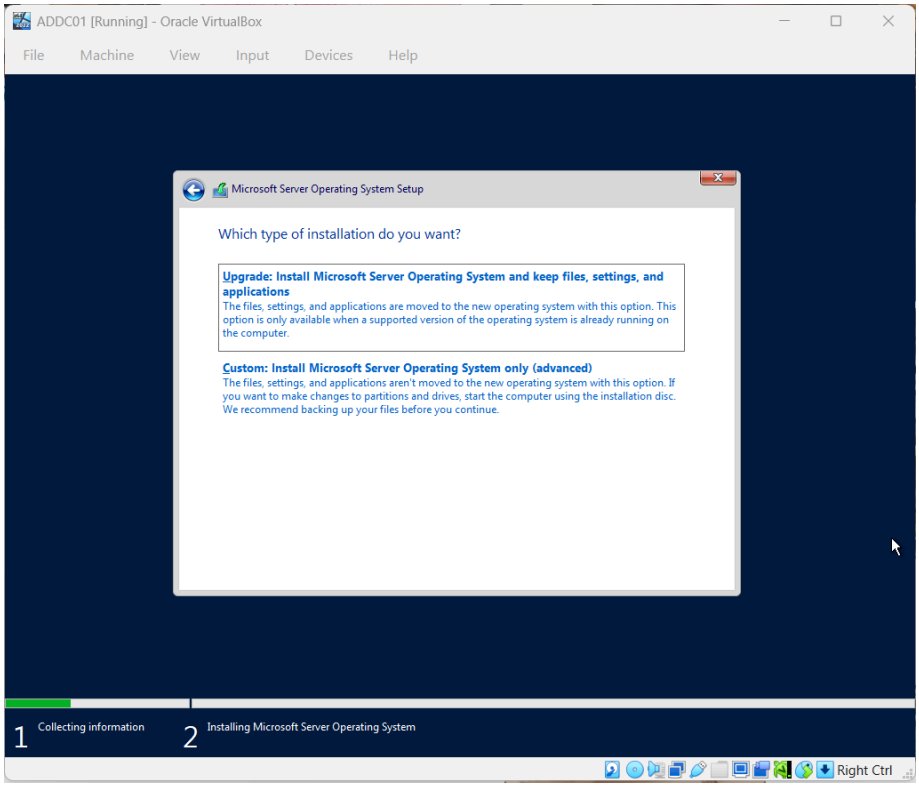
Clicked **Install Now**



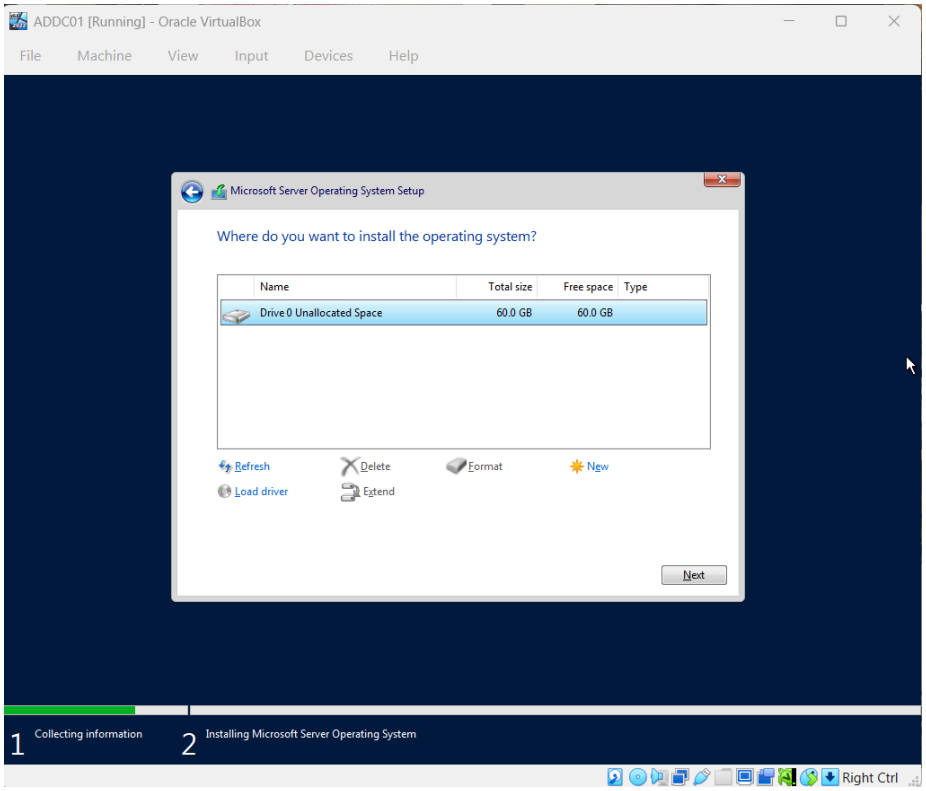
Clicked on *Windows Server 2022 Standard Evaluation Desktop Experience* and then hit **Next**



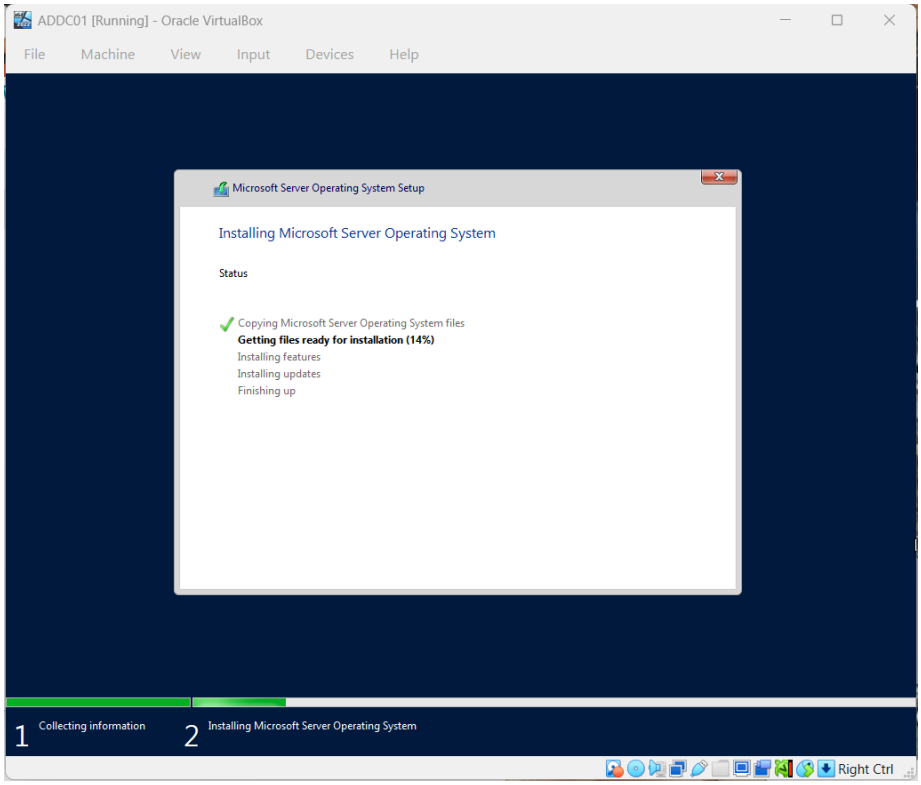
Read and Accepted the Terms and Conditions and pressed **Next**. Afterwards, I pressed Custom



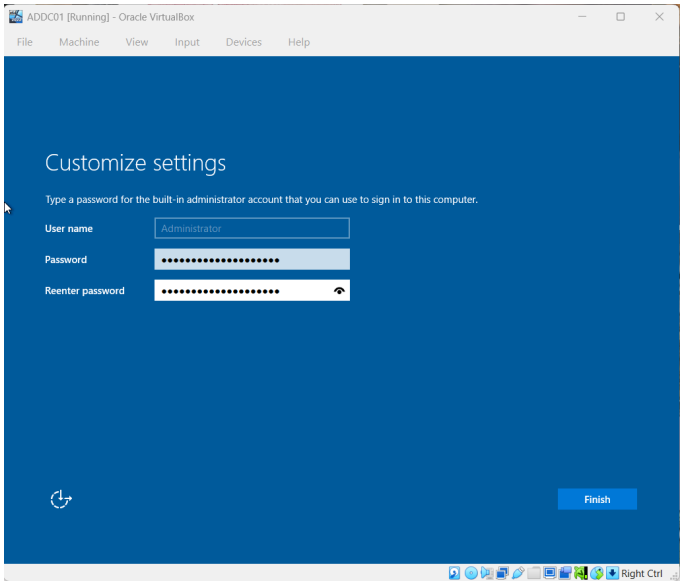
Clicked **Next** again



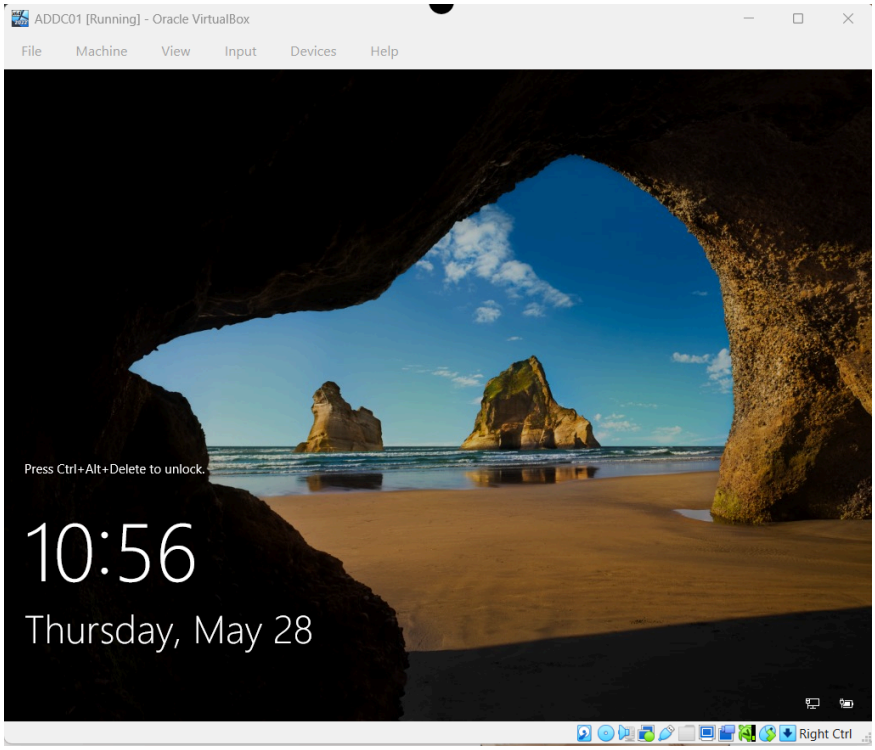
Then I came across this page that was *Installing Microsoft Server OS*

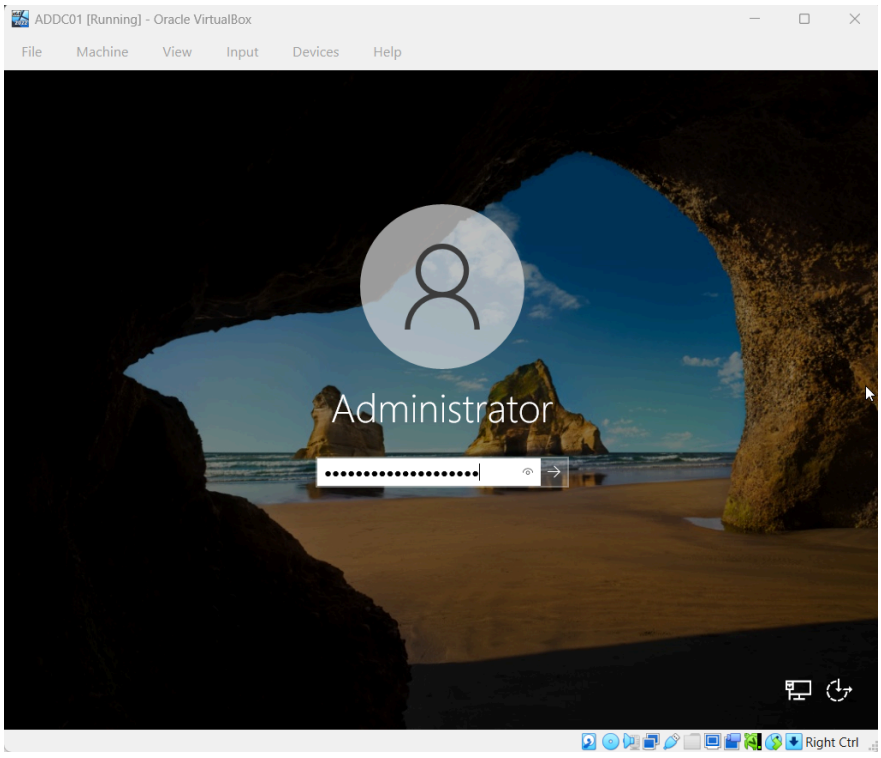


After a few minutes, this screen popped up to add a password for Admin log in. After adding the password I clicked **Finish**

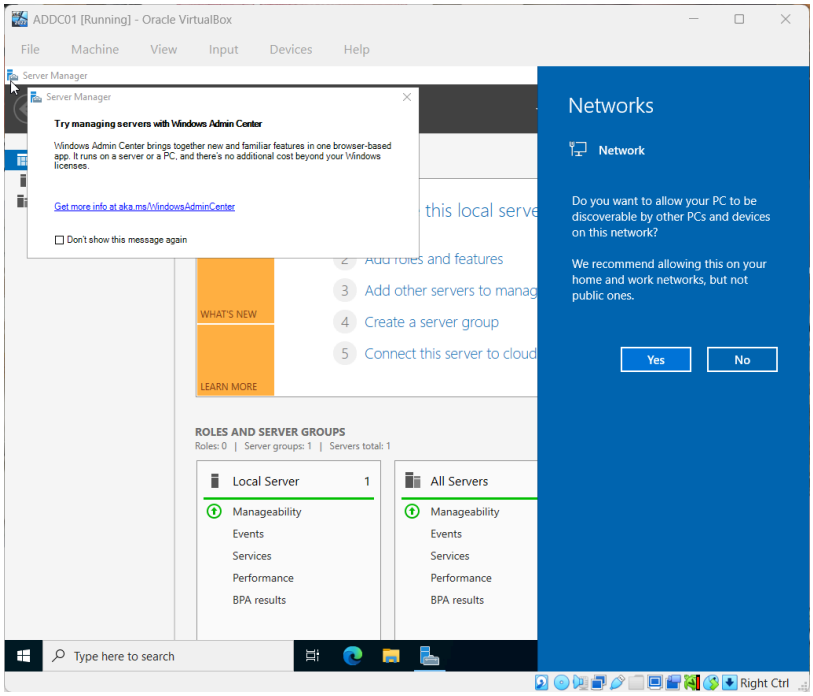


The log in screen appears shortly after, it prompted me to click control + alt + delete to unlock. I achieved this by going to **Input > Keyboard > Insert Ctrl-Alt-Del**

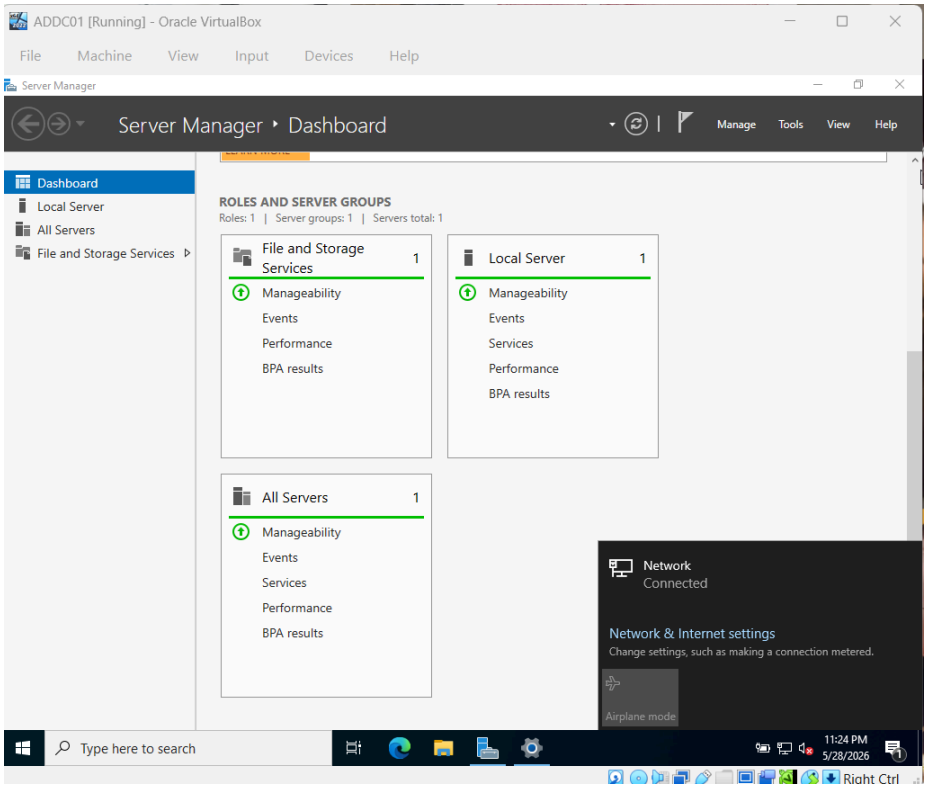




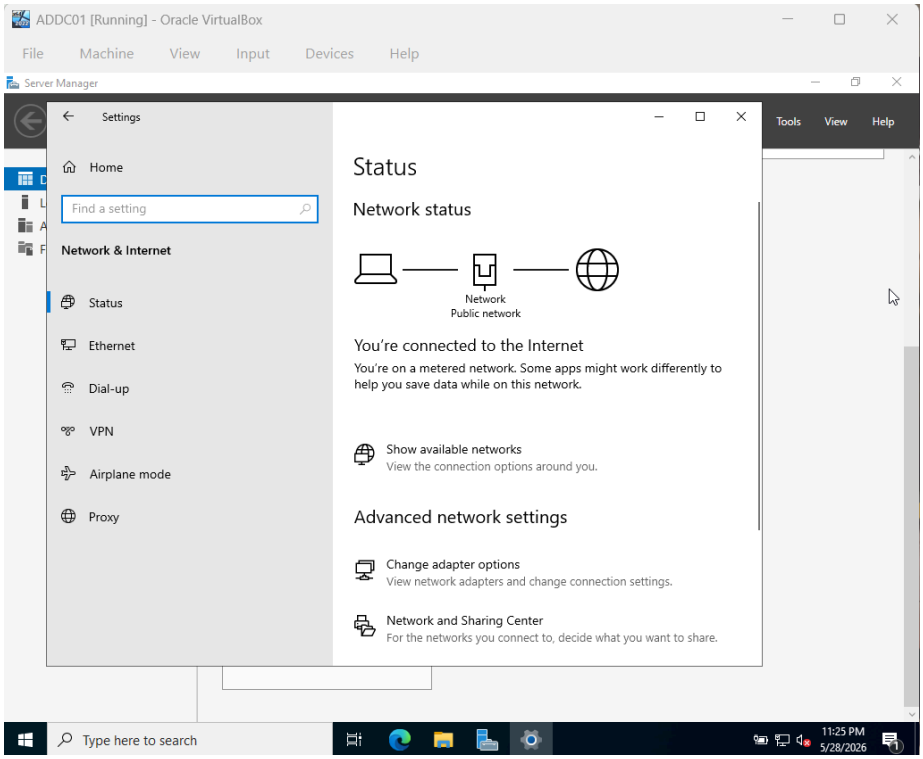
Once I'm logged in, Server Manager along with a popup that wanted to know if I wanted to make my PC discoverable. I clicked **No** and asked the small Server Manager window out.



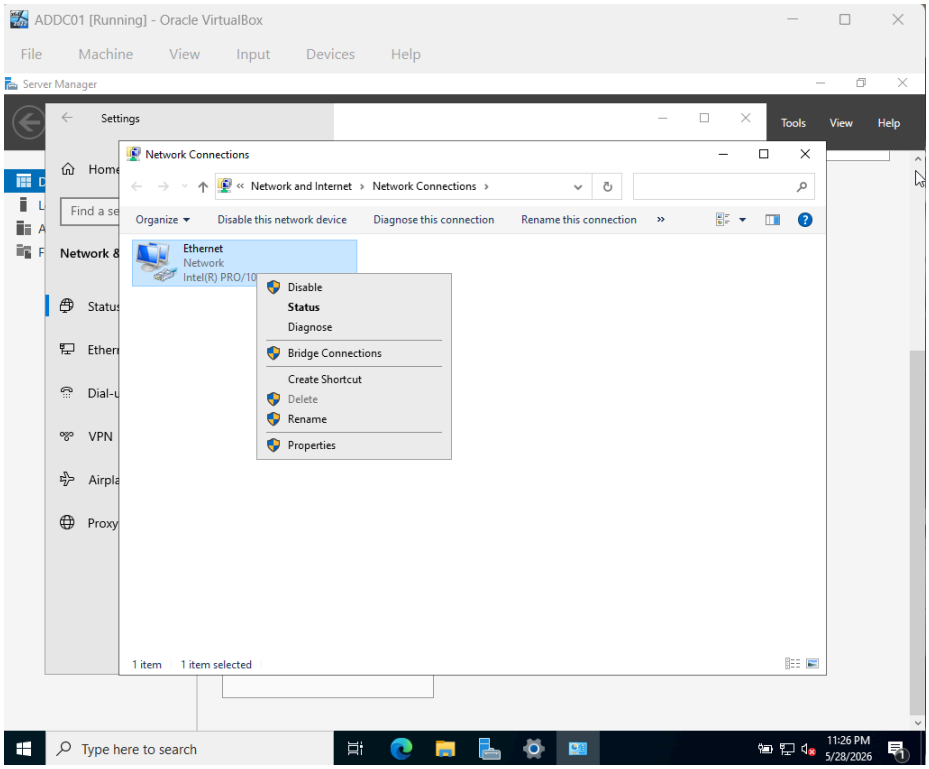
I decided the next step I wanted to take was to create the static IP. Scrolled all the way to the bottom to the Network icon and clicked on that. Clicked on **Network & Internet Settings**



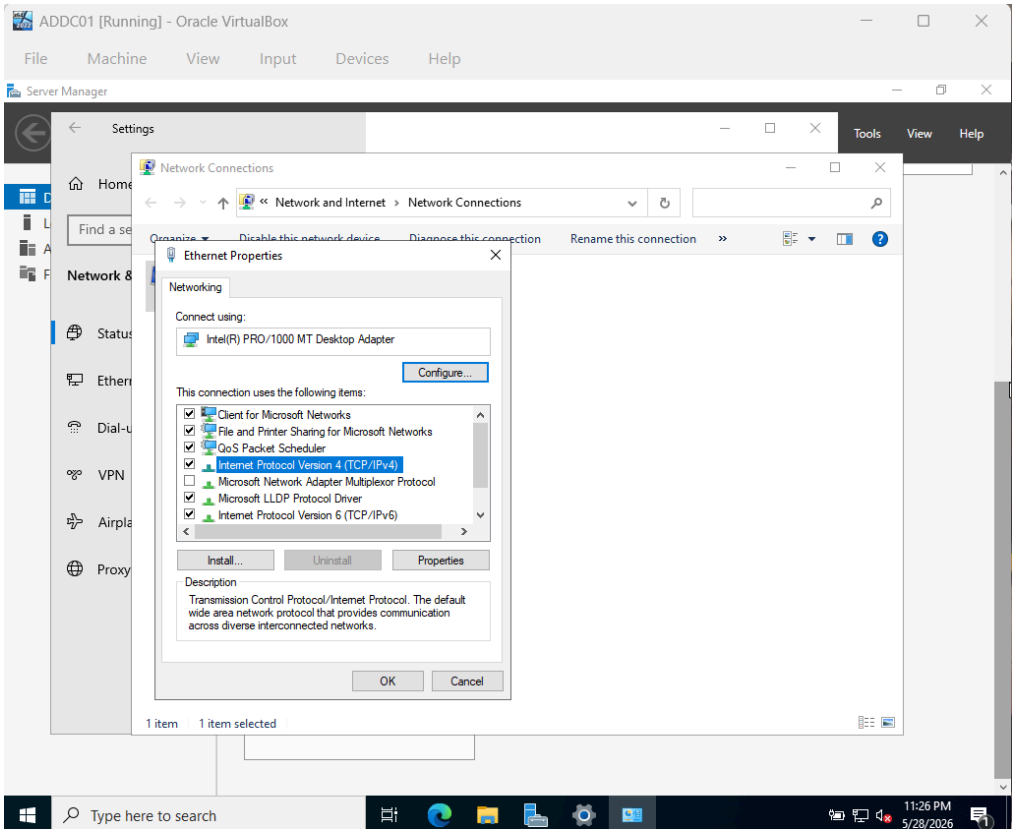
Clicked *Change adapter options* towards the bottom.

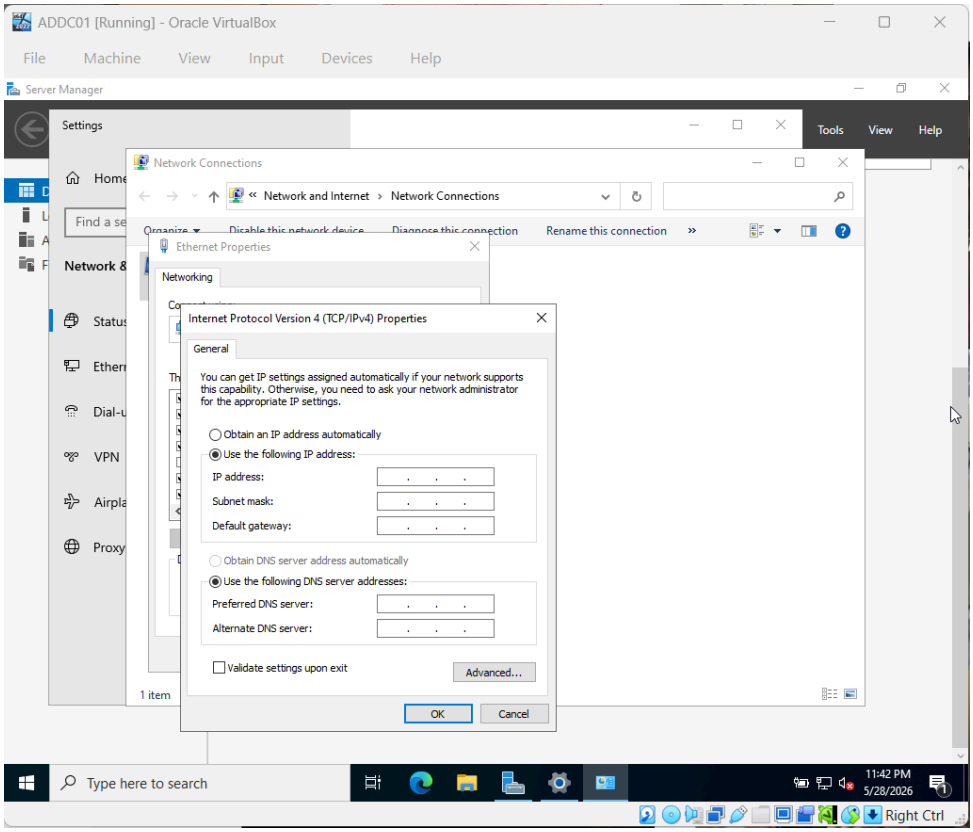


Right clicked on ethernet and clicked **Properties**.

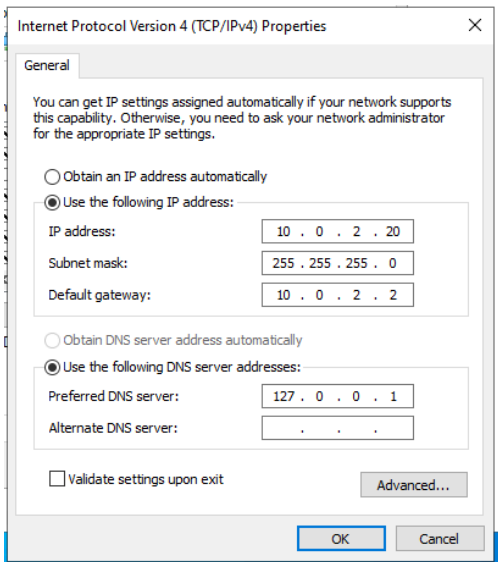


Clicked on *Internet Protocol Version 4 (TCP/IPv4)* and clicked **Properties**





I knew for the new IP address, I had to change only the 4th octet. For the Subnet Mask and Default Gateway, I kept all octets the same as the ones on Windows 11 Pro. For the DNS I went with a preferred server 127.0.0.1. I clicked **OK**.



I went to the command prompt and typed in *ipconfig* to confirm the changes were made successfully.

```
C:\Users\Administrator>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : fd17:625c:f037:2:ad10:5ced:7c7d:573b
   Link-local IPv6 Address . . . . . : fe80::ad10:5ced:7c7d:573b%11
   IPv4 Address. . . . . . . . . . . : 10.0.2.20
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::2%11
                                       10.0.2.2

C:\Users\Administrator>
```

I also pinged the static IP to make sure the network adapter accepted the settings

```
C:\Users\Administrator>ping 10.0.2.20

Pinging 10.0.2.20 with 32 bytes of data:
Reply from 10.0.2.20: bytes=32 time<1ms TTL=128
Reply from 10.0.2.20: bytes=32 time=1ms TTL=128
Reply from 10.0.2.20: bytes=32 time<1ms TTL=128
Reply from 10.0.2.20: bytes=32 time<1ms TTL=128
```

I also pinged the LocalHost, Windows 11 VM (10.0.2.15), and the router/gateway (10.0.2.2)

```
C:\Users\Administrator>ping localhost

Pinging ADDC01 [::1] with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time=2ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms

Ping statistics for ::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

```
C:\Users\Administrator>ping 10.0.2.2

Pinging 10.0.2.2 with 32 bytes of data:
Reply from 10.0.2.2: bytes=32 time=1ms TTL=255
Reply from 10.0.2.2: bytes=32 time=1ms TTL=255
Reply from 10.0.2.2: bytes=32 time=4ms TTL=255
Reply from 10.0.2.2: bytes=32 time=1ms TTL=255

Ping statistics for 10.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 1ms
```

The ping to Windows failed. At first I mistakenly typed the IP address on its own. Which is why the result was a “not recognized” error. The second attempt resulted in “Destination host unreachable”. I was able to resolve this issue by enabling NAT Network on both VMs.

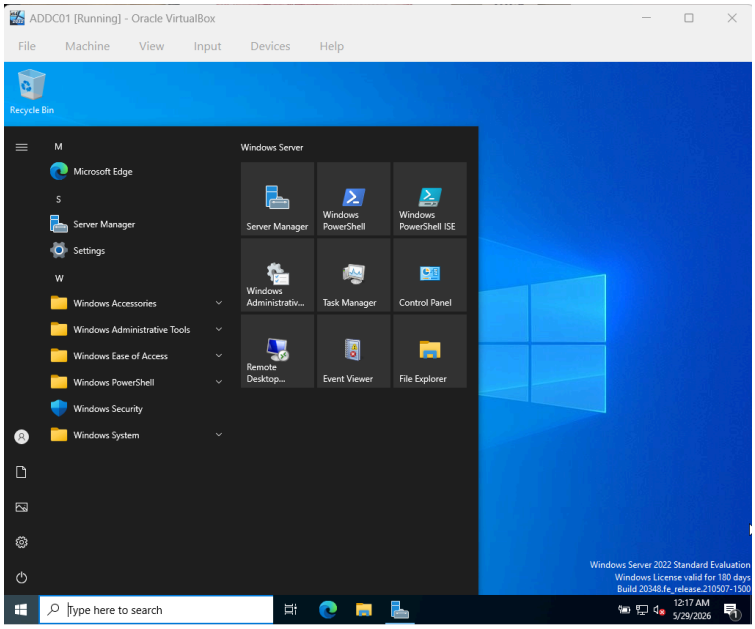
```
C:\Users\Administrator>10.0.2.15
'10.0.2.15' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Administrator>ping 10.0.2.15

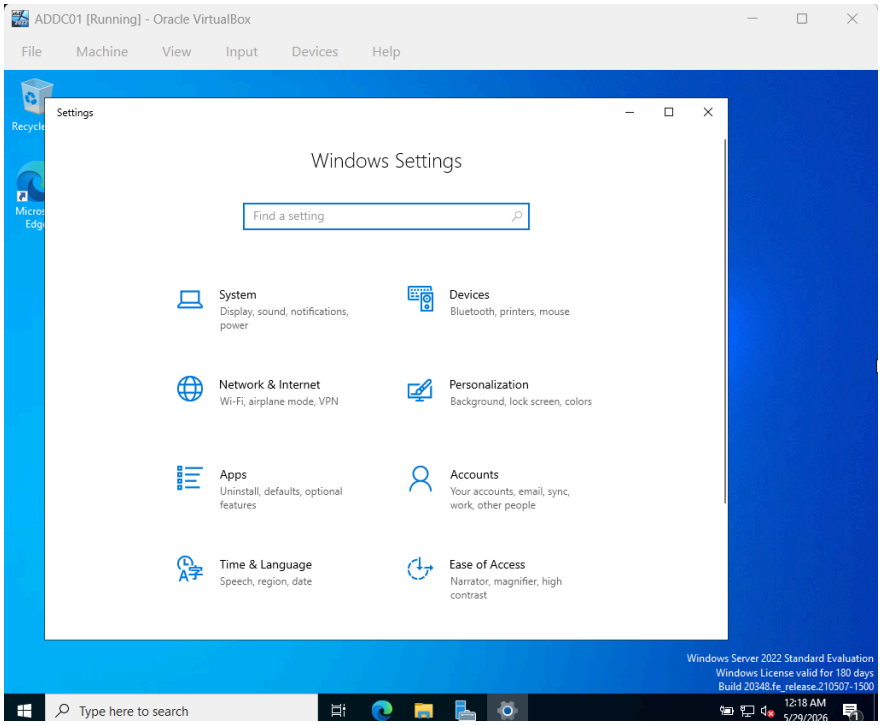
Pinging 10.0.2.15 with 32 bytes of data:
Reply from 10.0.2.20: Destination host unreachable.
Reply from 10.0.2.20: Destination host unreachable.
Reply from 10.0.2.20: Destination host unreachable.
Reply from 10.0.2.20: Destination host unreachable.

Ping statistics for 10.0.2.15:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

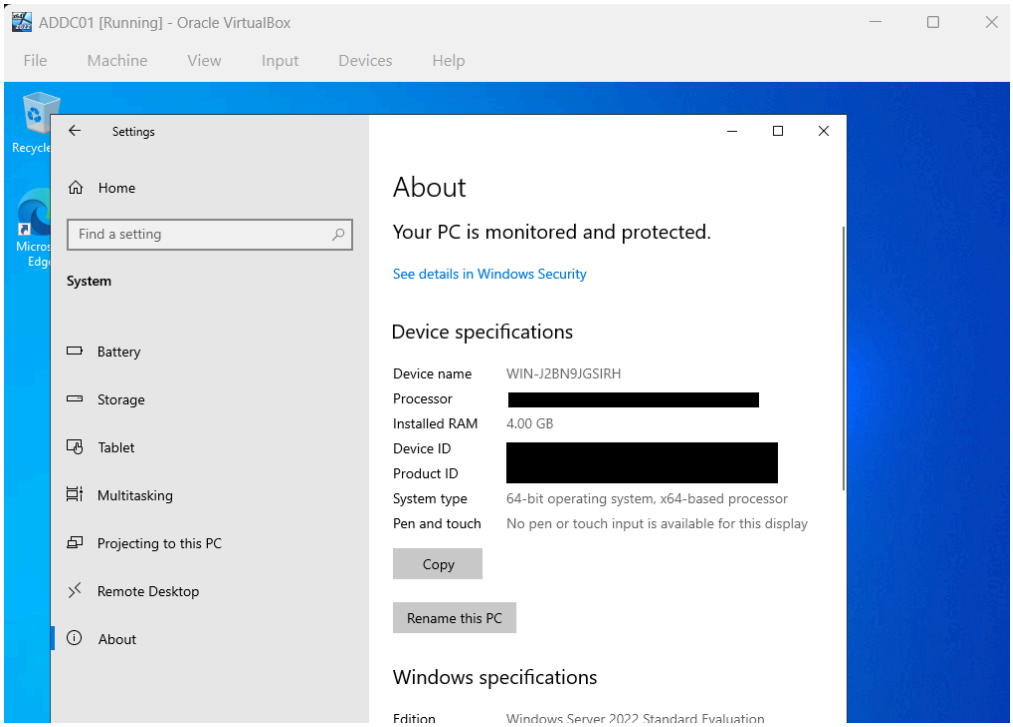
After I created a static IP, I changed the name of the server. I first went to the Window icon and clicked settings.



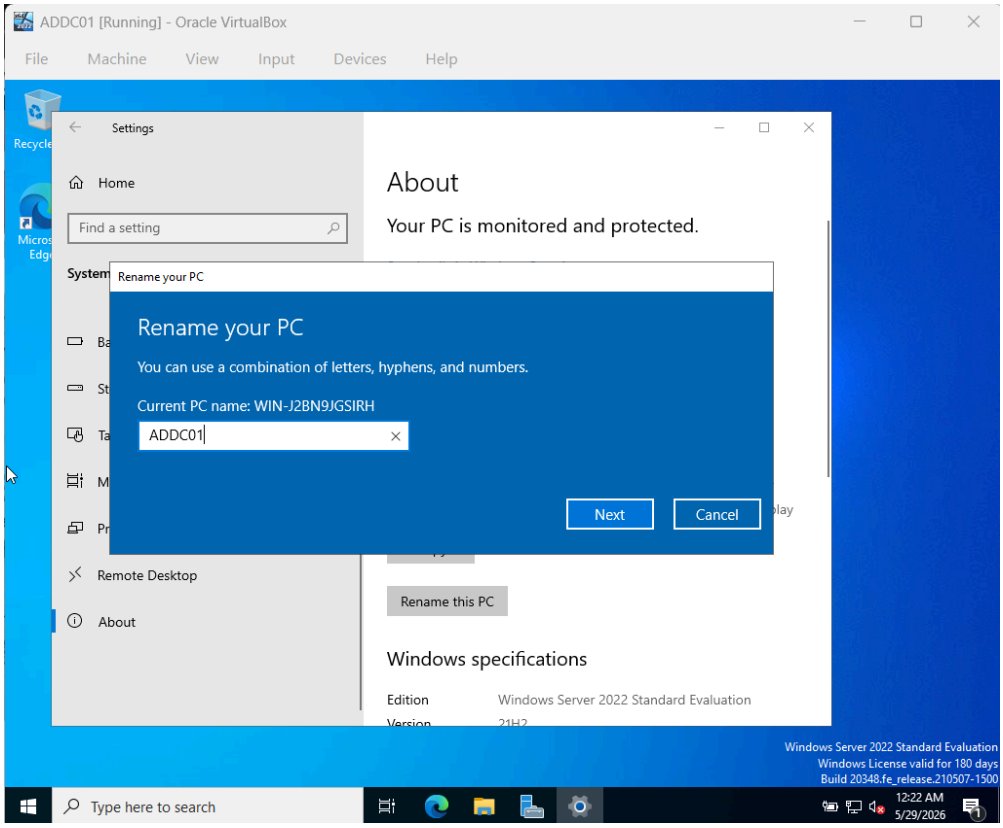
Then went to System



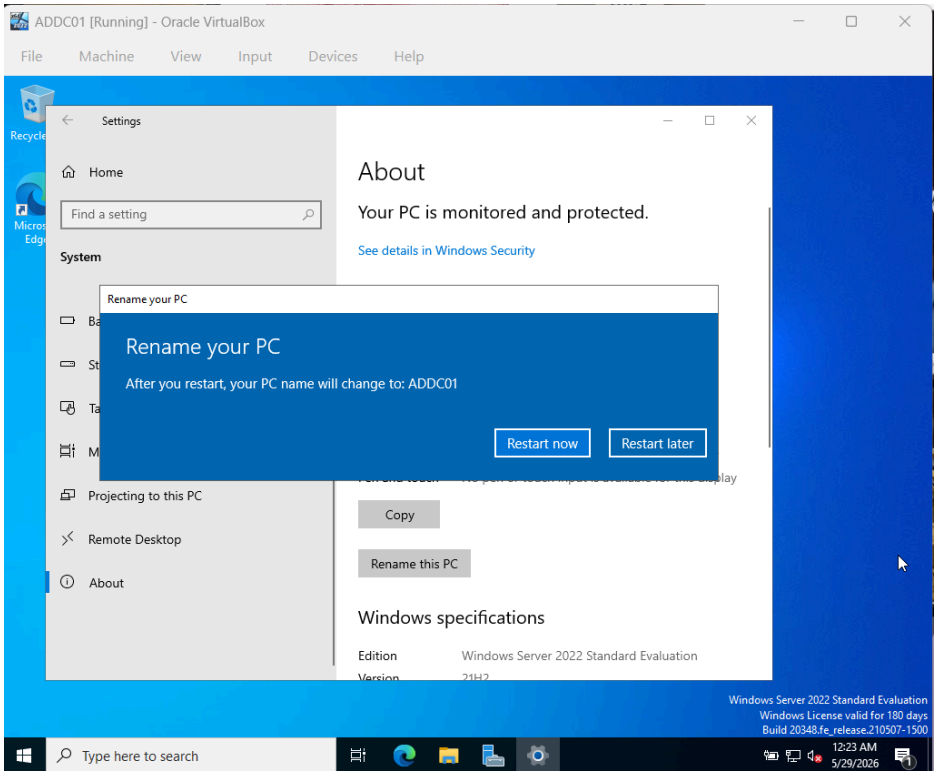
Hit **About > Rename this PC**



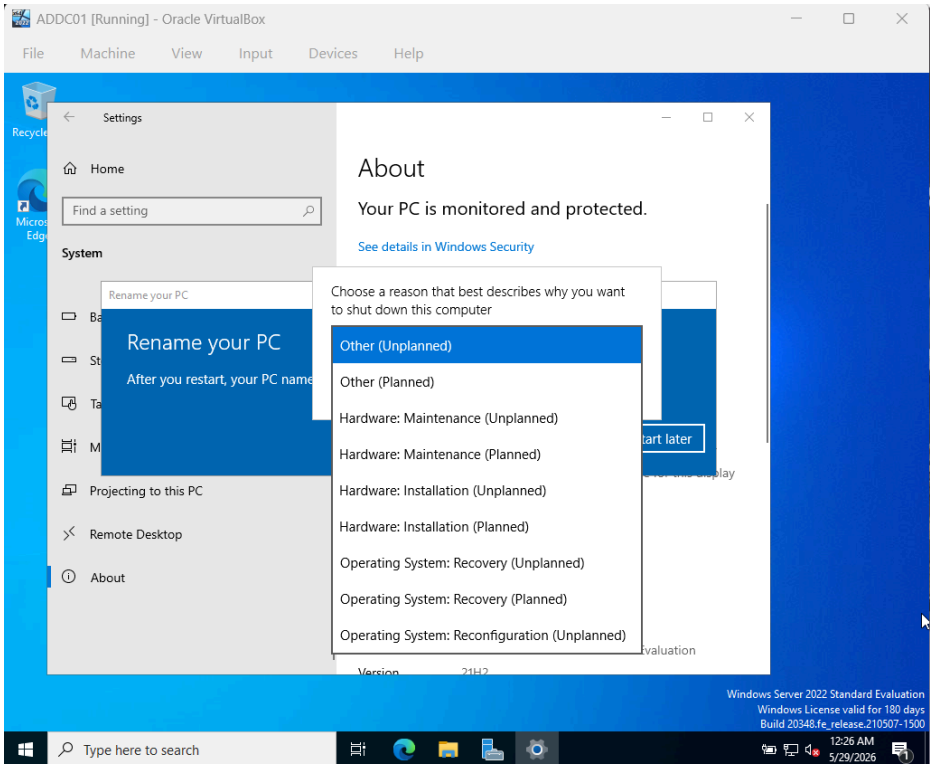
Created the new name and hit **Next**.



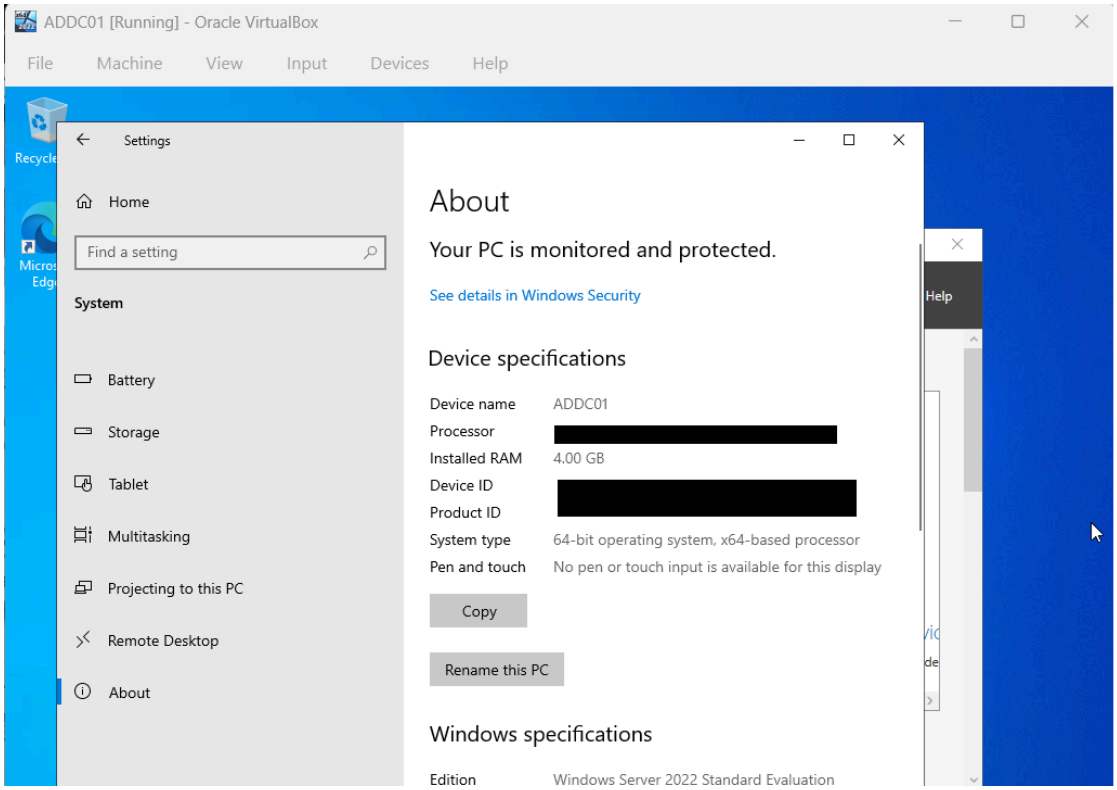
Prompted to restart now shortly after and I clicked **Restart now**



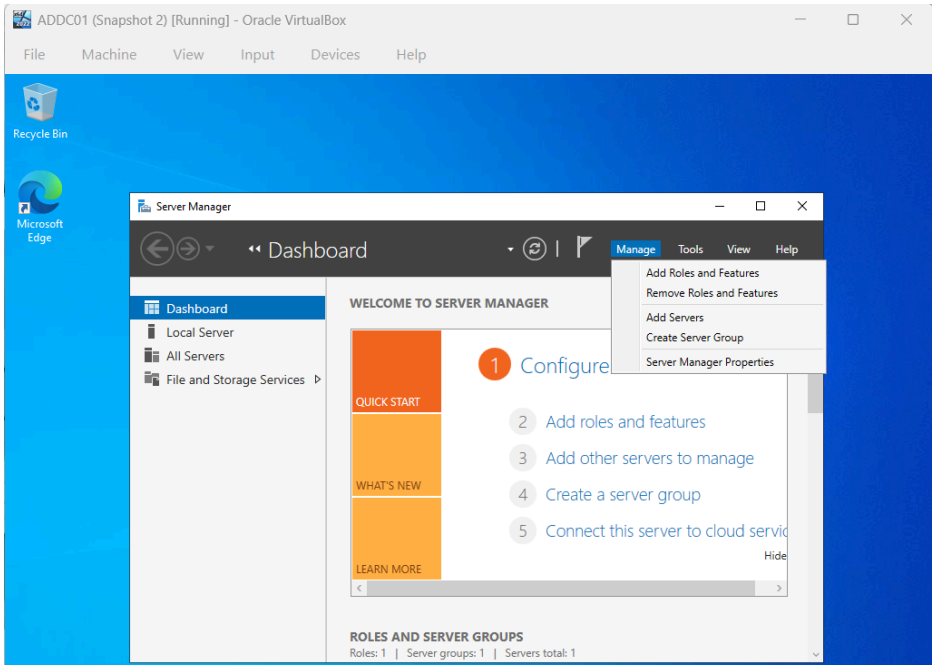
Clicked *Other (Unplanned)* and pressed continue.



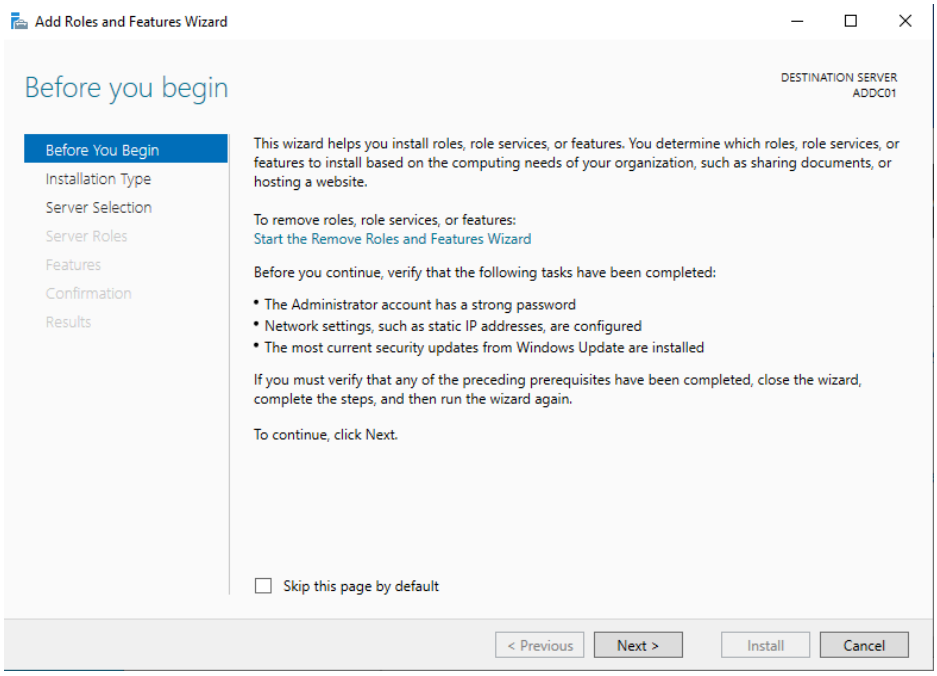
Logged back into the system and went back to where I renamed the PC to ensure that the changes were properly updated.



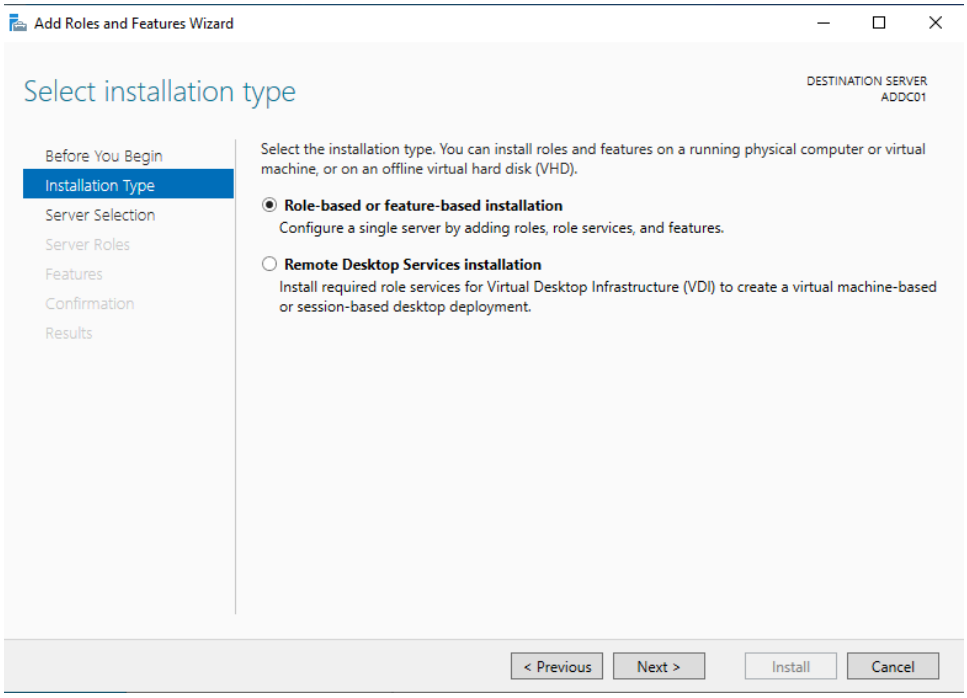
Once I verified this step was complete, I installed the active directory domain services to the Windows Server. I did this by first going to *Server Manager* > *Manage* > *Add Roles and Features*



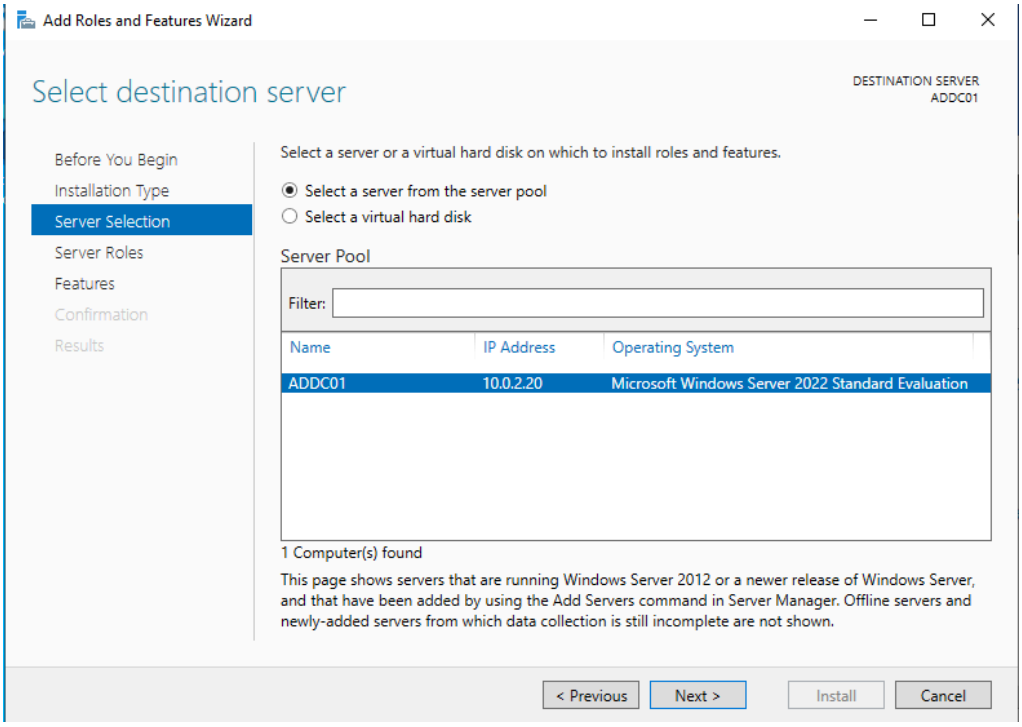
Clicked made sure all of the following tasks were complete and then hit **Next**



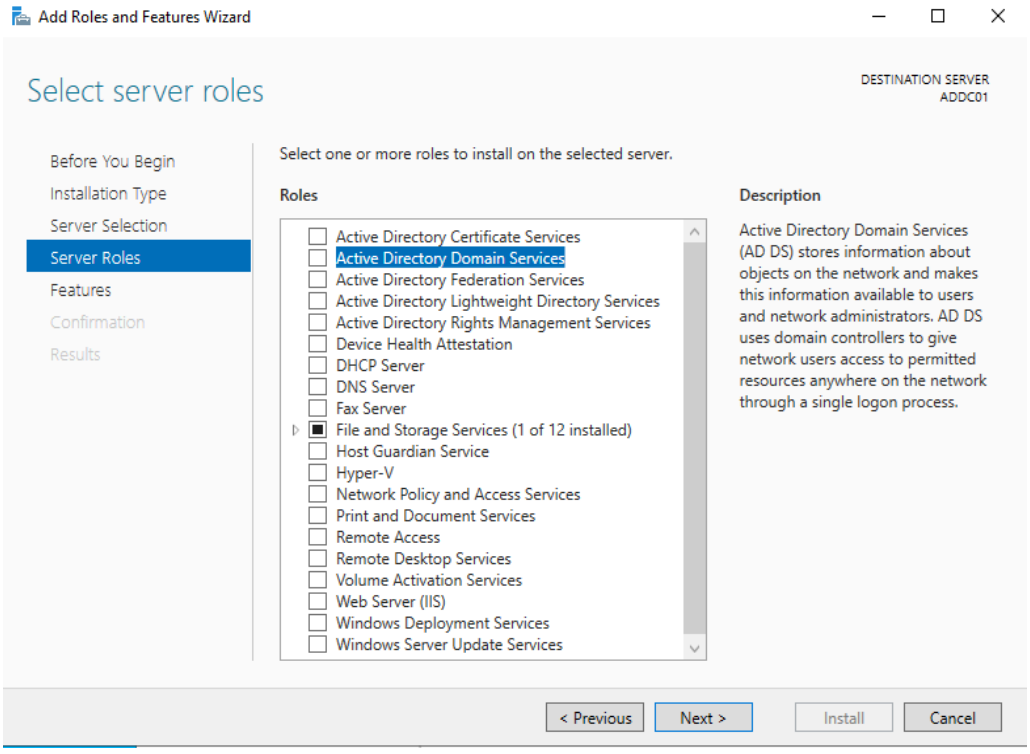
Kept the setting on *Role-based of feature-based installation* and clicked **Next**



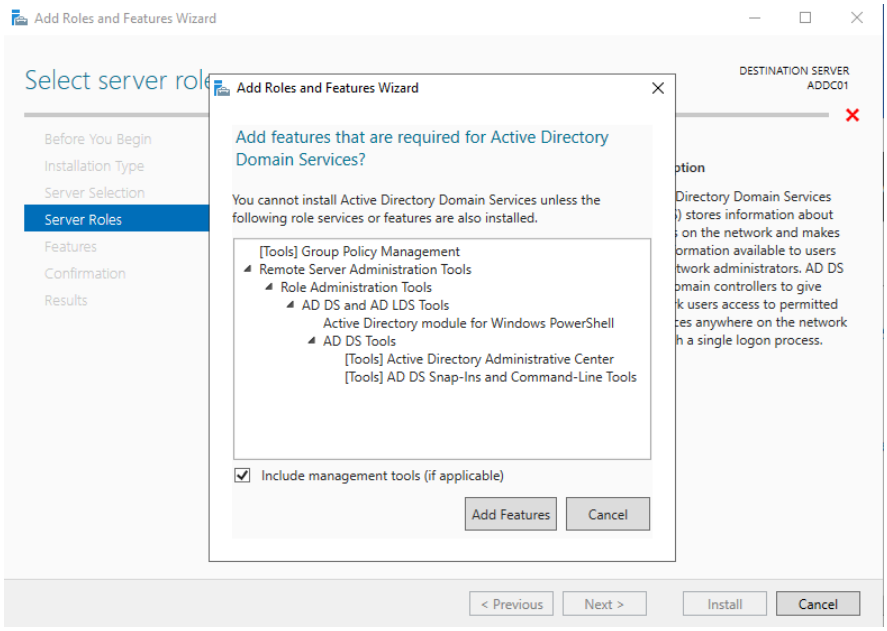
I only have one server which is ADDC01 and clicked **Next**



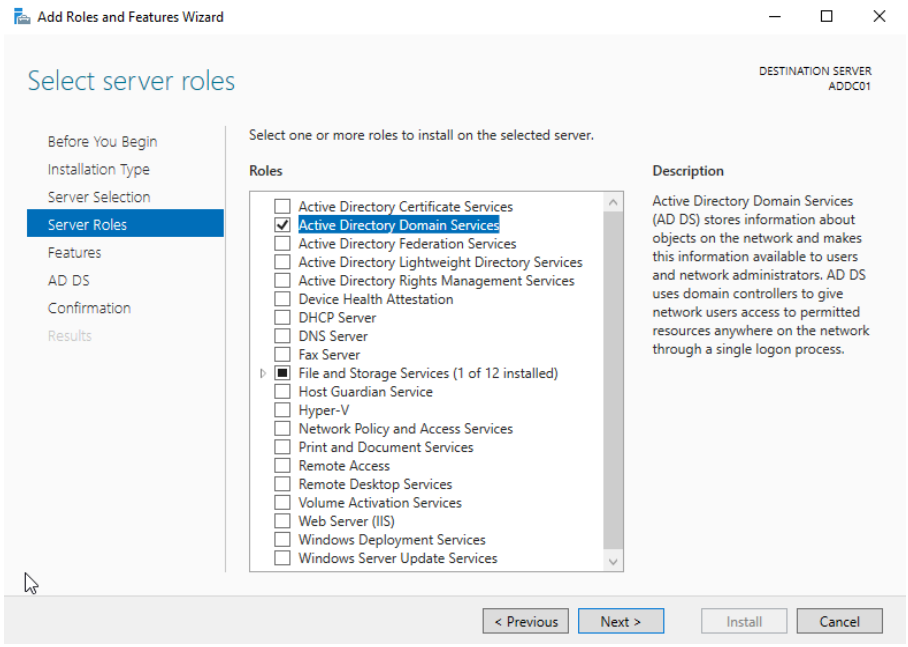
Clicked on *Active Director Domain Services*



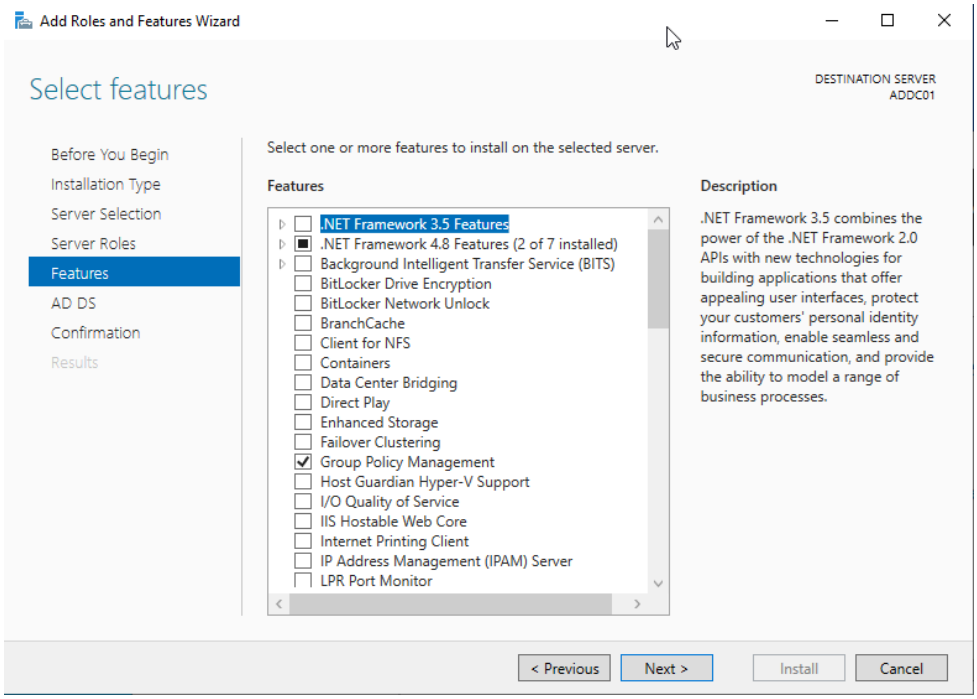
Clicked **Add Features**



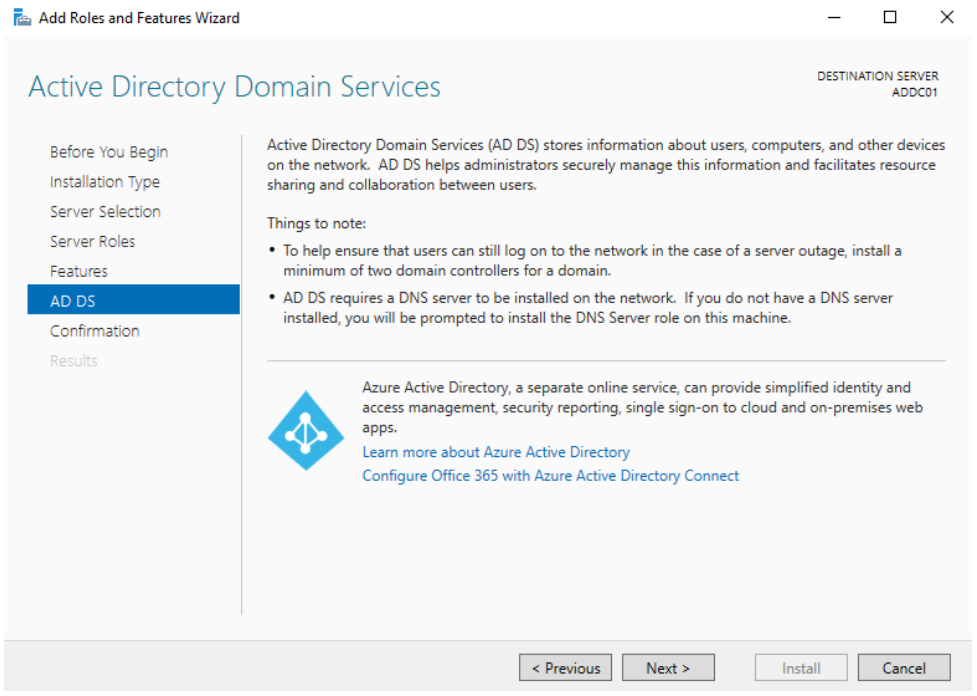
Clicked **Next**



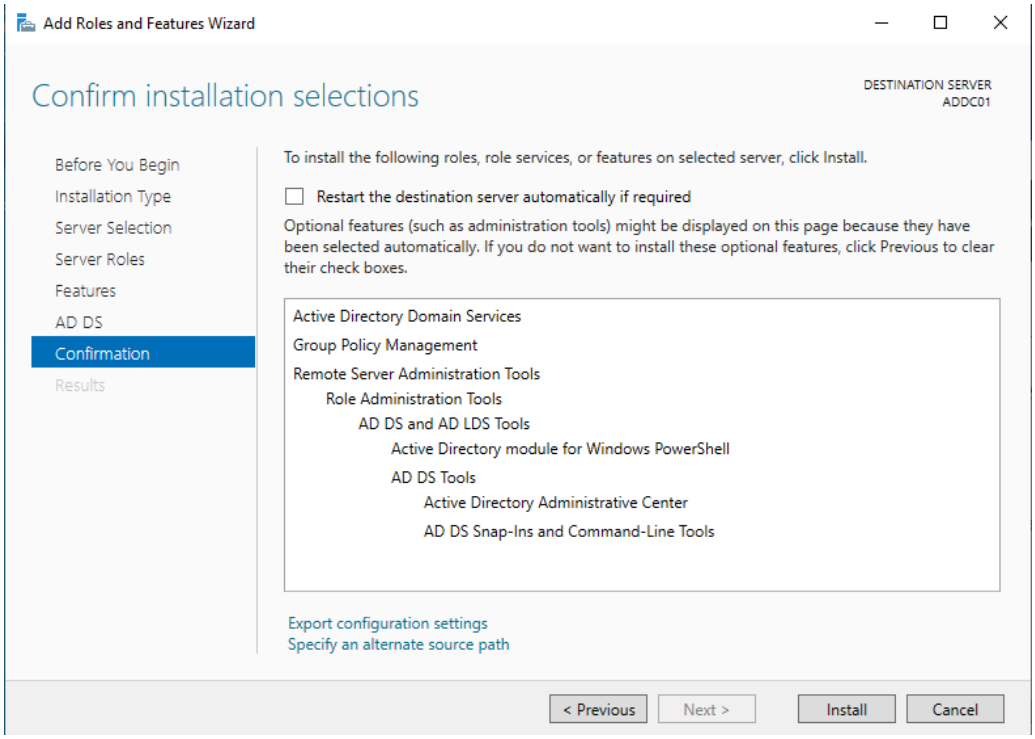
Clicked **Next**

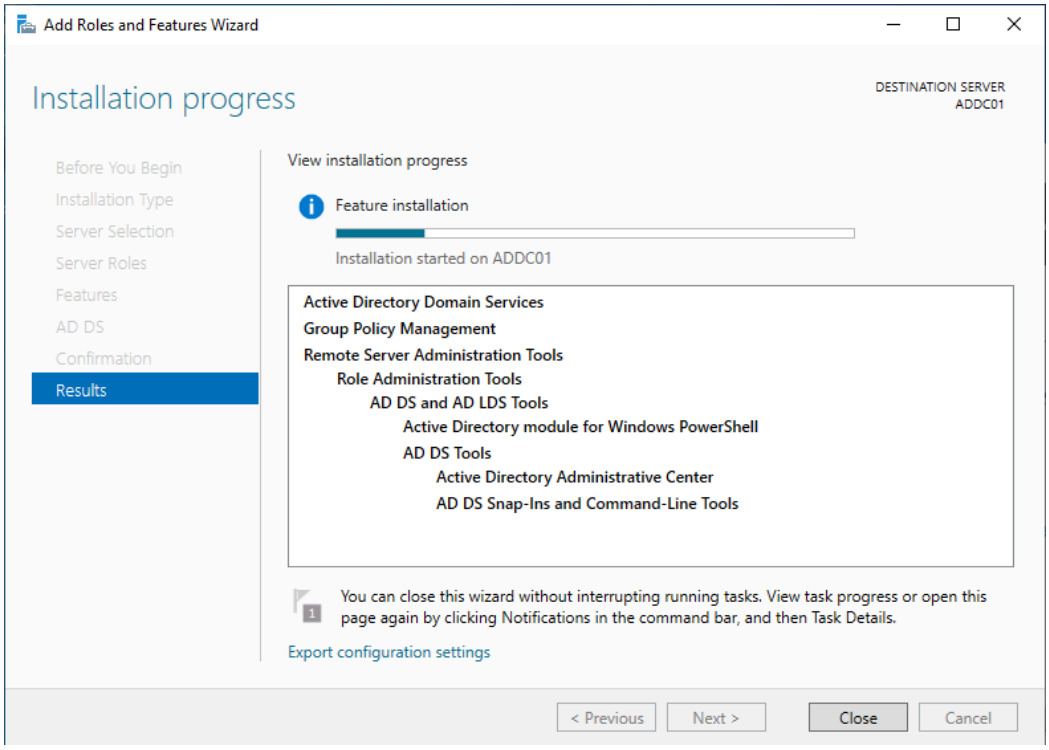


Read and clicked **Next**

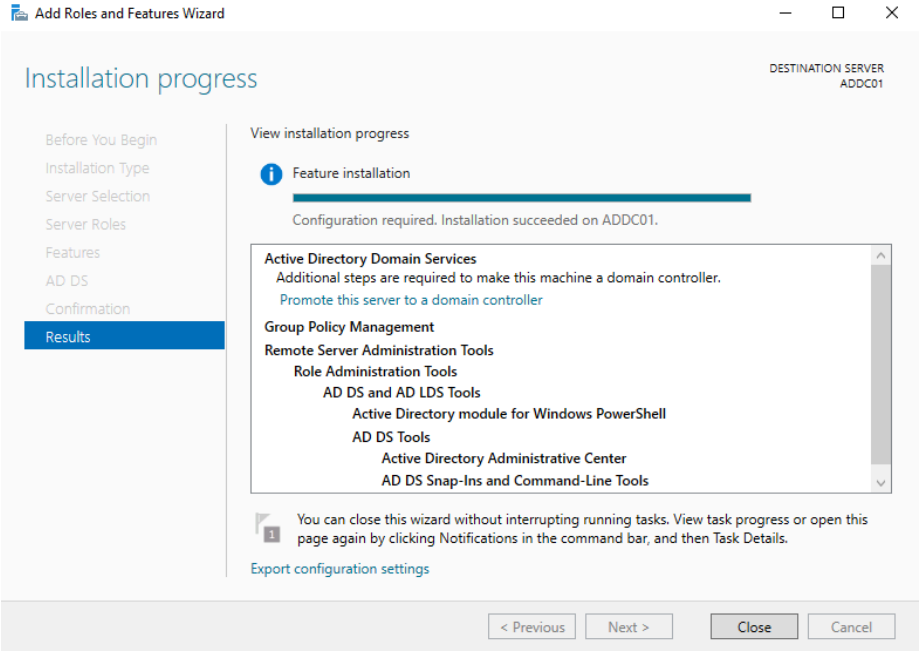


Clicked **Install**

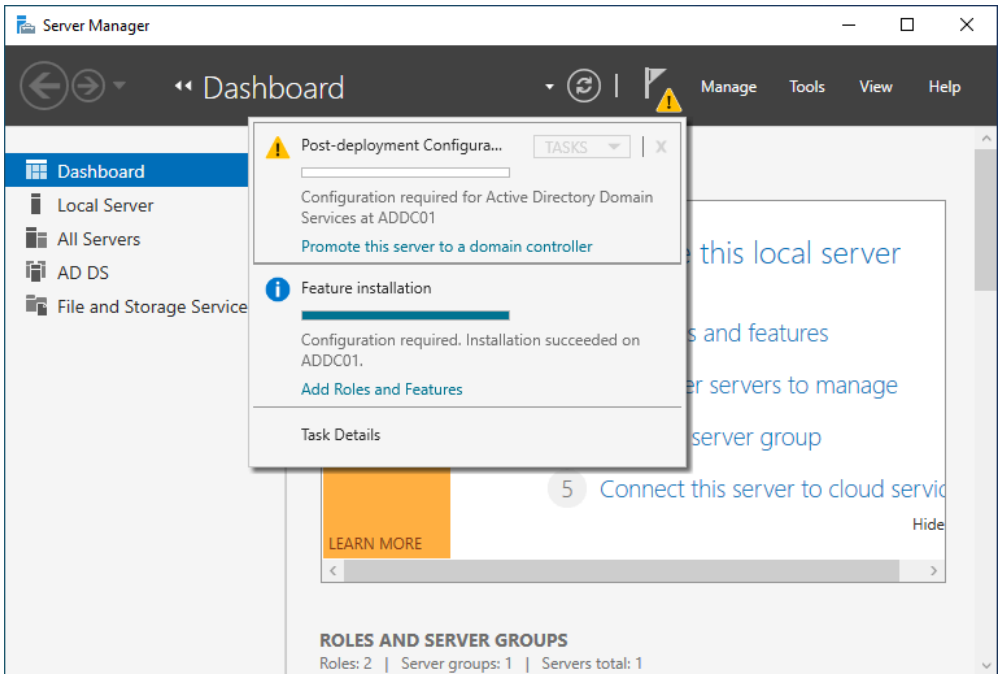




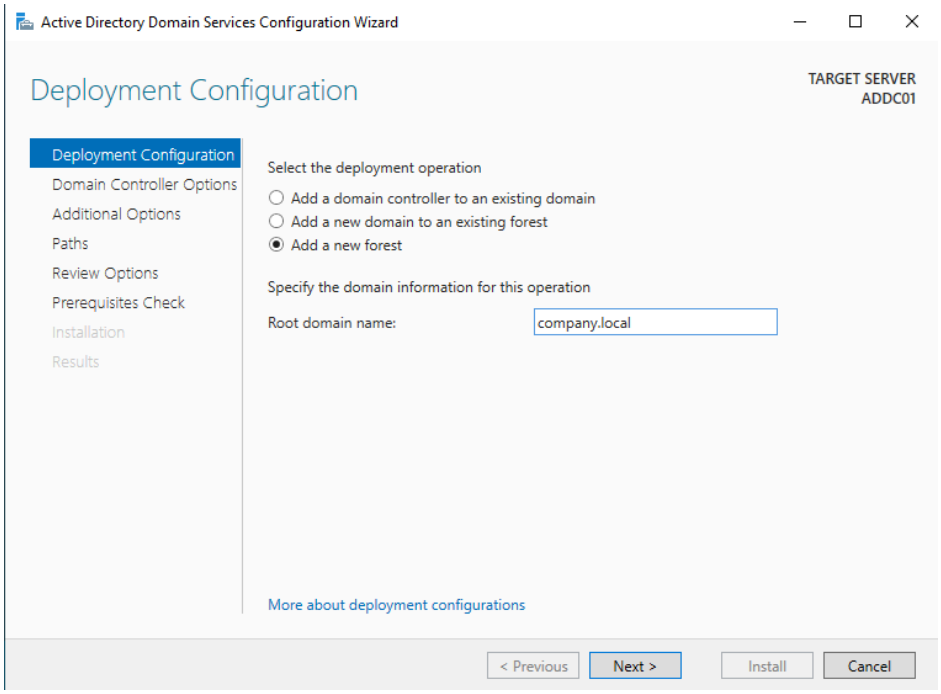
I closed out by clicking **Close** once I saw the installation succeeded.



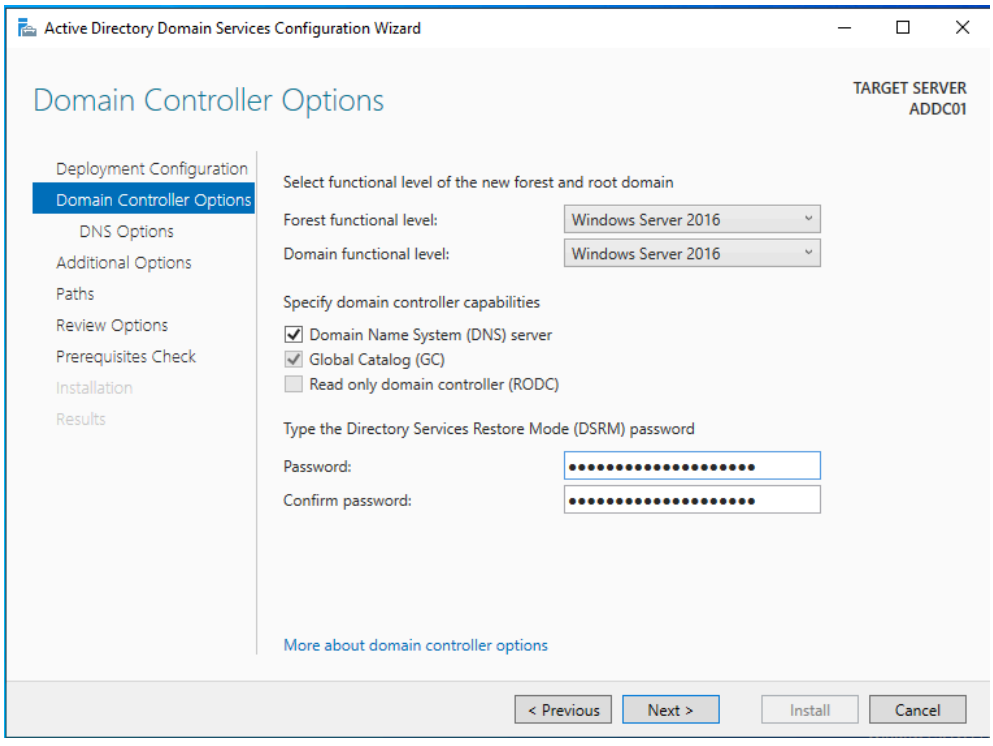
Went back to Server Manager and clicked on the Flag icon and pressed on *Promote this server to a domain controller.*



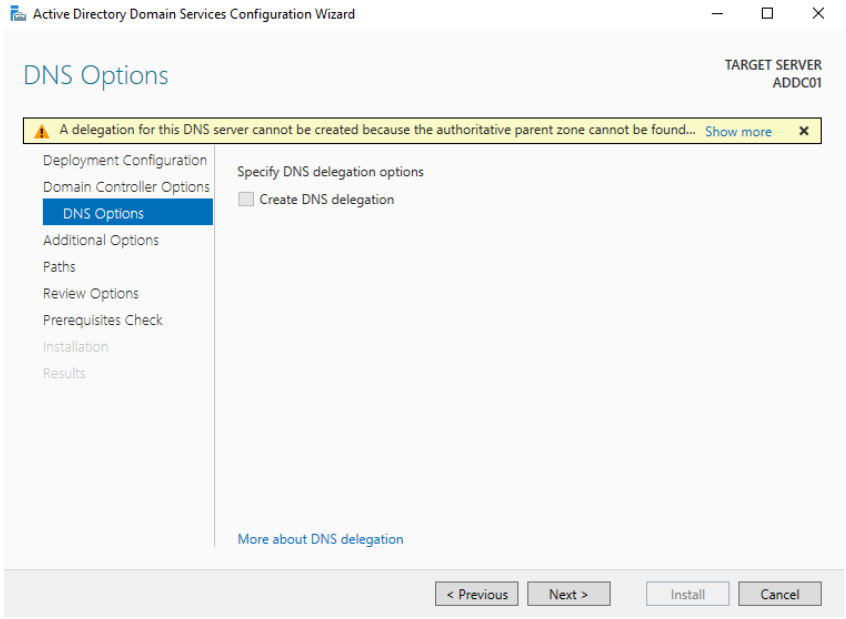
I clicked on *Add a new forest* and typed in company.local for the Root domain name and hit **Next**



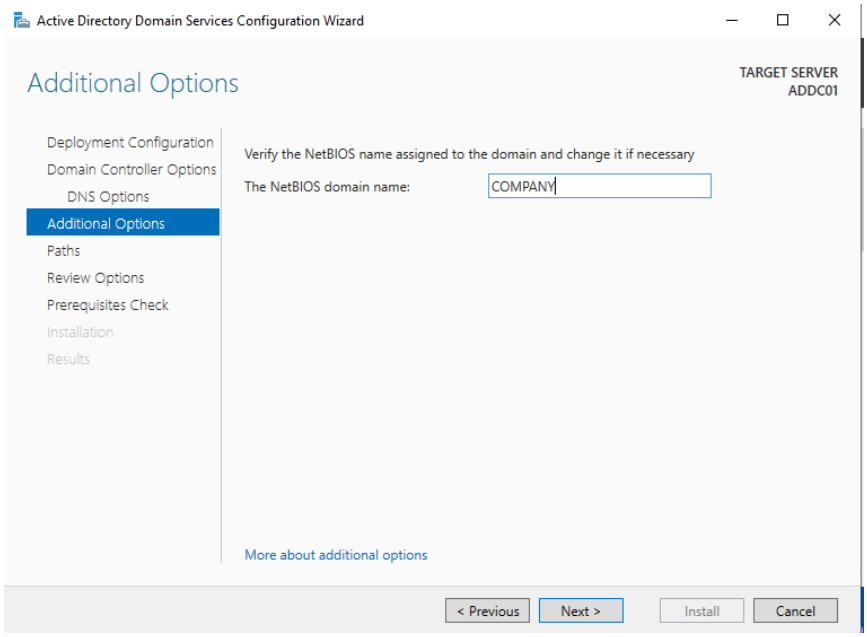
Created a strong password and hit **Next**



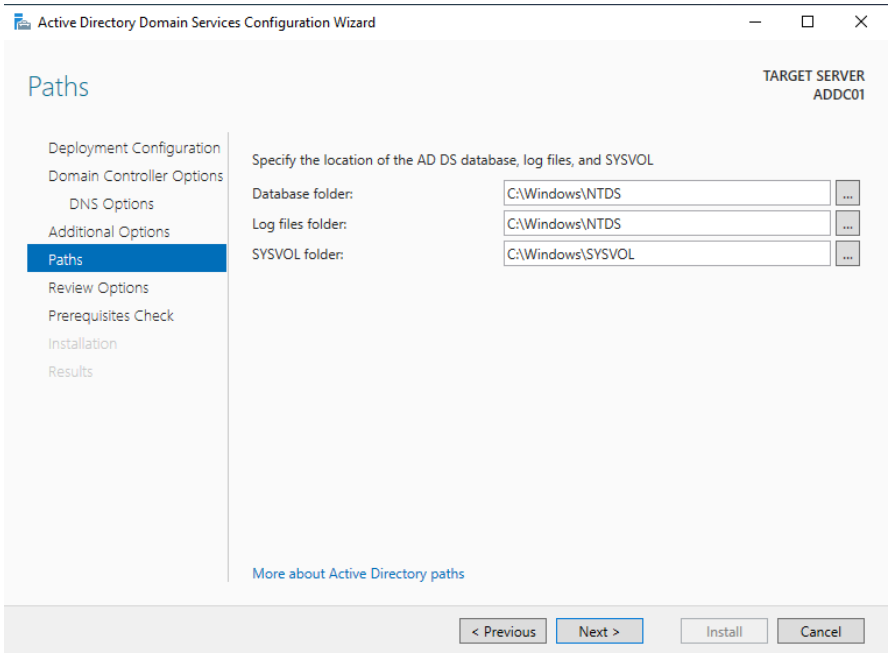
Hit **Next**



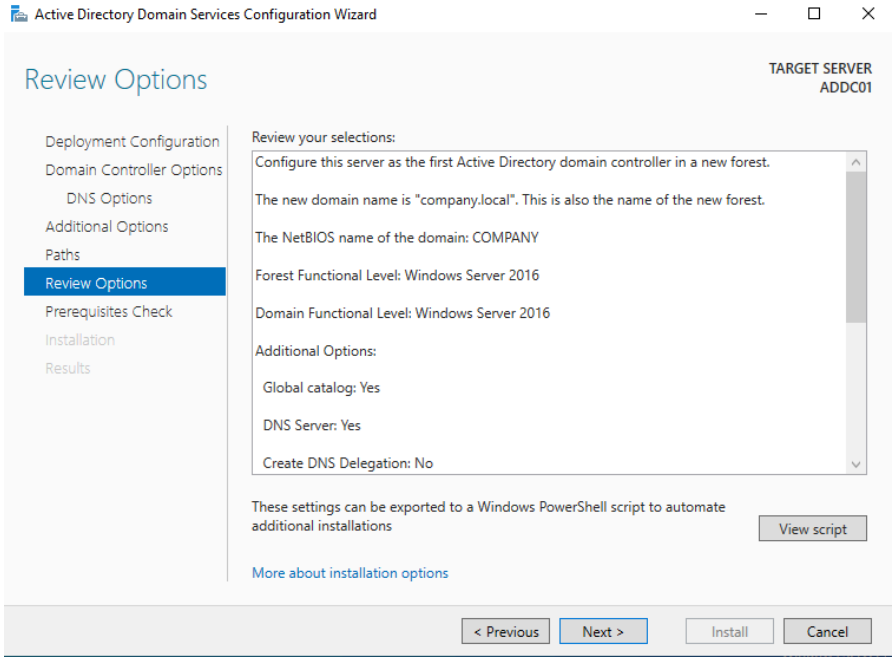
Clicked **Next**



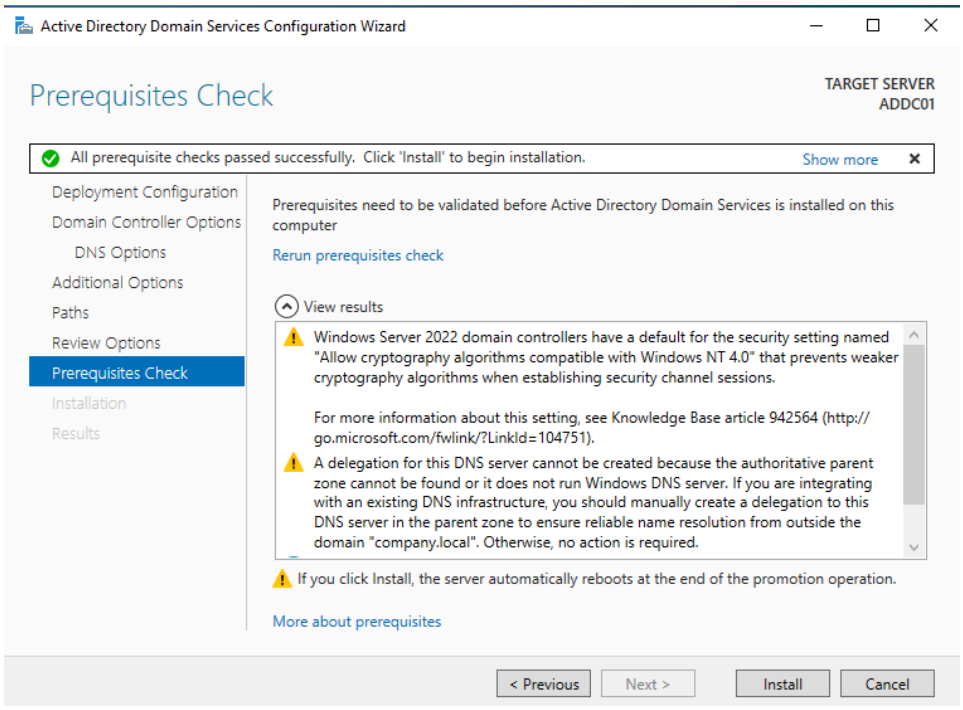
Click **Next**

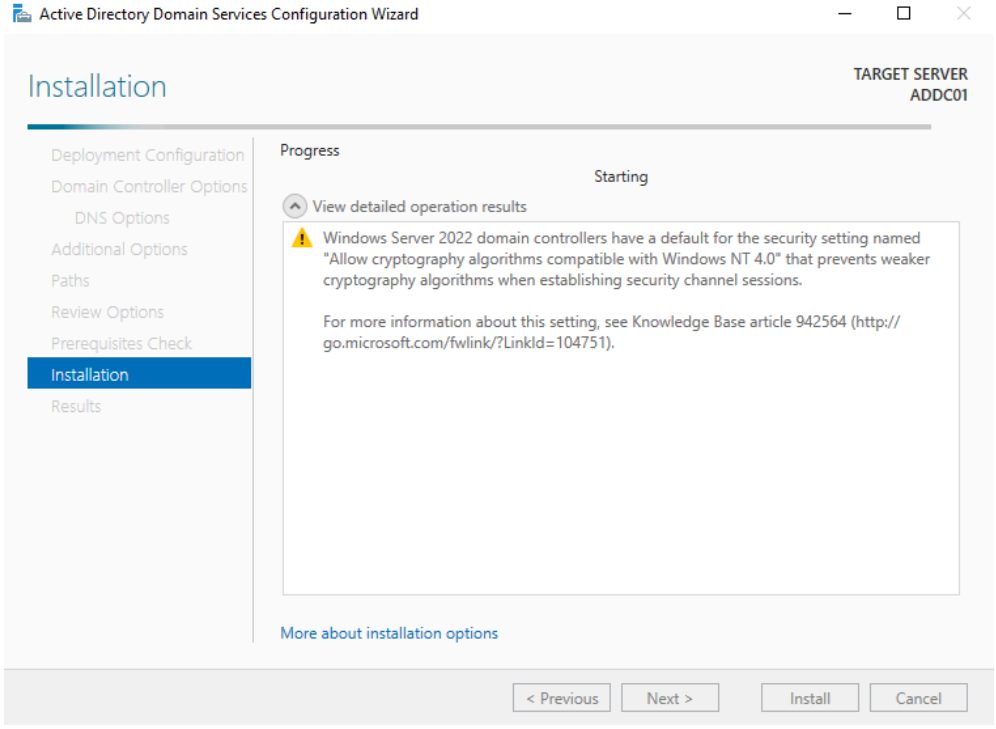


I read and then clicked **Next**

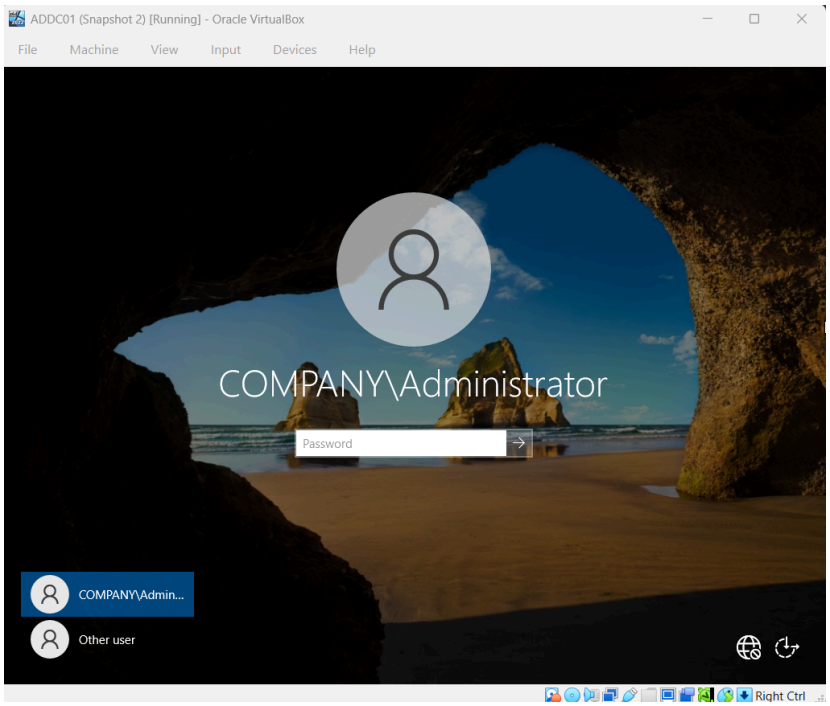


Hit **Install**

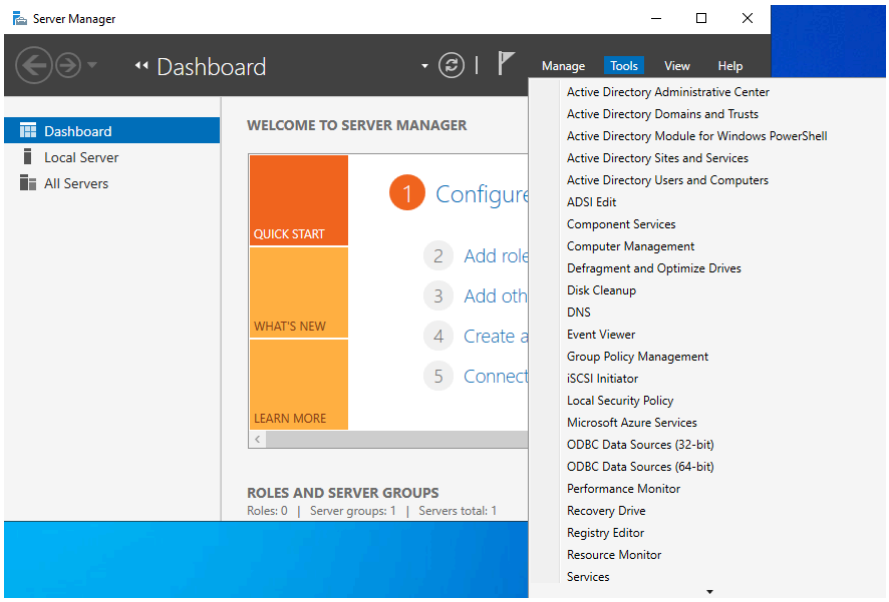

Active Directory Domain Services Configuration Wizard
Prerequisites Check
TARGET SERVER
ADDC01
All prerequisite checks passed successfully. Click 'Install' to begin installation.
Show more
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Prerequisites need to be validated before Active Directory Domain Services is installed on this computer
Rerun prerequisites check
View results
Windows Server 2022 domain controllers have a default for the security setting named "Allow cryptography algorithms compatible with Windows NT 4.0" that prevents weaker cryptography algorithms when establishing security channel sessions.
For more information about this setting, see Knowledge Base article 942564 (http://go.microsoft.com/fwlink/?LinkId=104751).
A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found or it does not run Windows DNS server. If you are integrating with an existing DNS infrastructure, you should manually create a delegation to this DNS server in the parent zone to ensure reliable name resolution from outside the domain "company.local". Otherwise, no action is required.
If you click Install, the server automatically reboots at the end of the promotion operation.
More about prerequisites
< Previous
Next >
Install
Cancel


Active Directory Domain Services Configuration Wizard
Installation
TARGET SERVER
ADDC01
Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results
Progress
Starting
View detailed operation results
Windows Server 2022 domain controllers have a default for the security setting named "Allow cryptography algorithms compatible with Windows NT 4.0" that prevents weaker cryptography algorithms when establishing security channel sessions.
For more information about this setting, see Knowledge Base article 942564 (http://go.microsoft.com/fwlink/?LinkId=104751).
More about installation options
< Previous
Next >
Install
Cancel

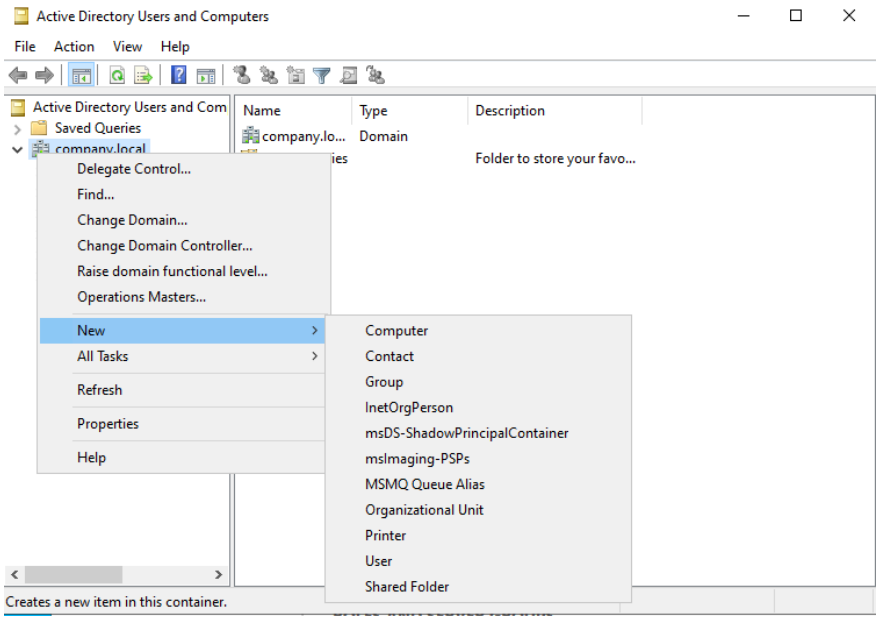
Once installed, it automatically restarted and the domain name appeared



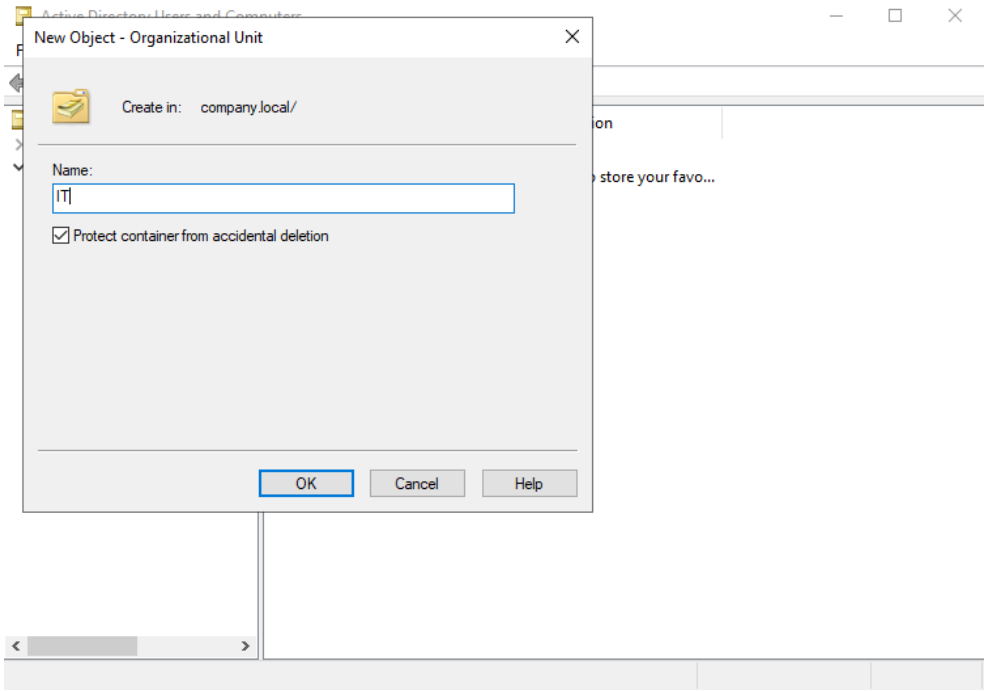
I logged in to create users. Went to *Server Manager > Tools > Active Directory Users and Computers*



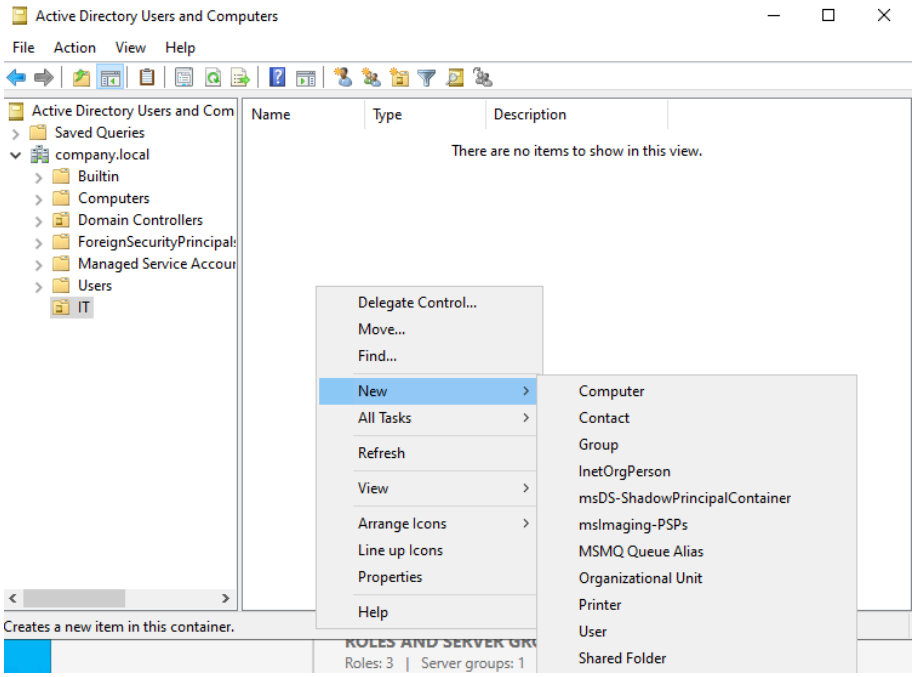
Once this window appeared, I right clicked on *company.local* > *new* > *Organizational Unit*



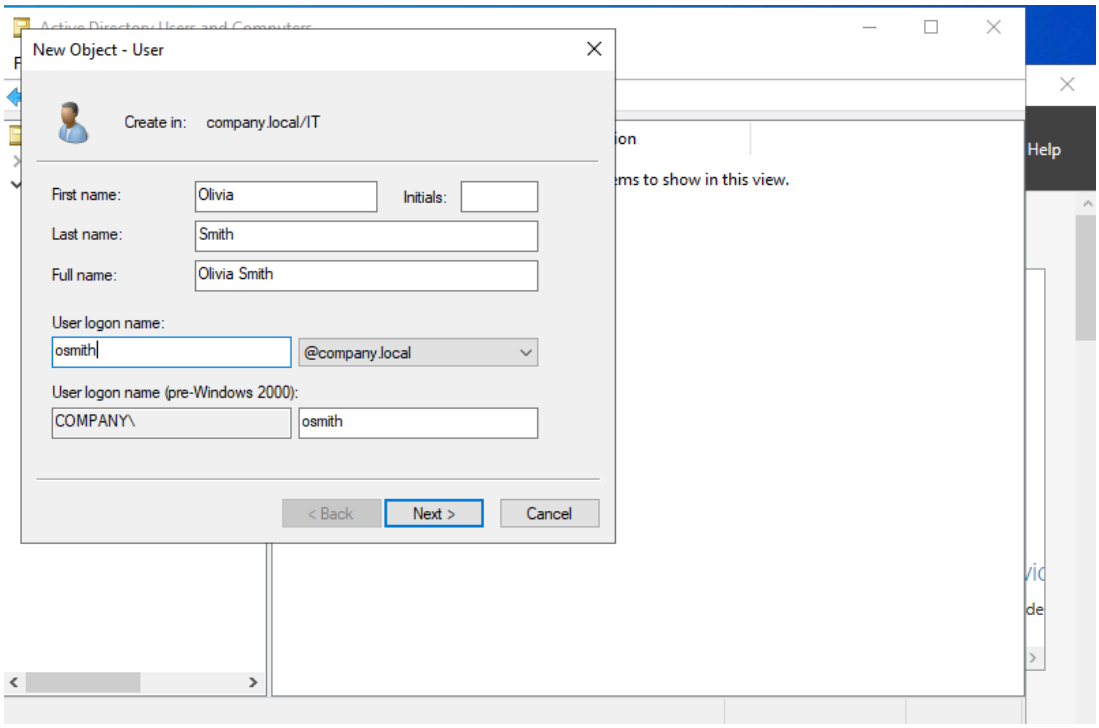
For one of the Organizational Units, I typed *IT* and pressed **OK**



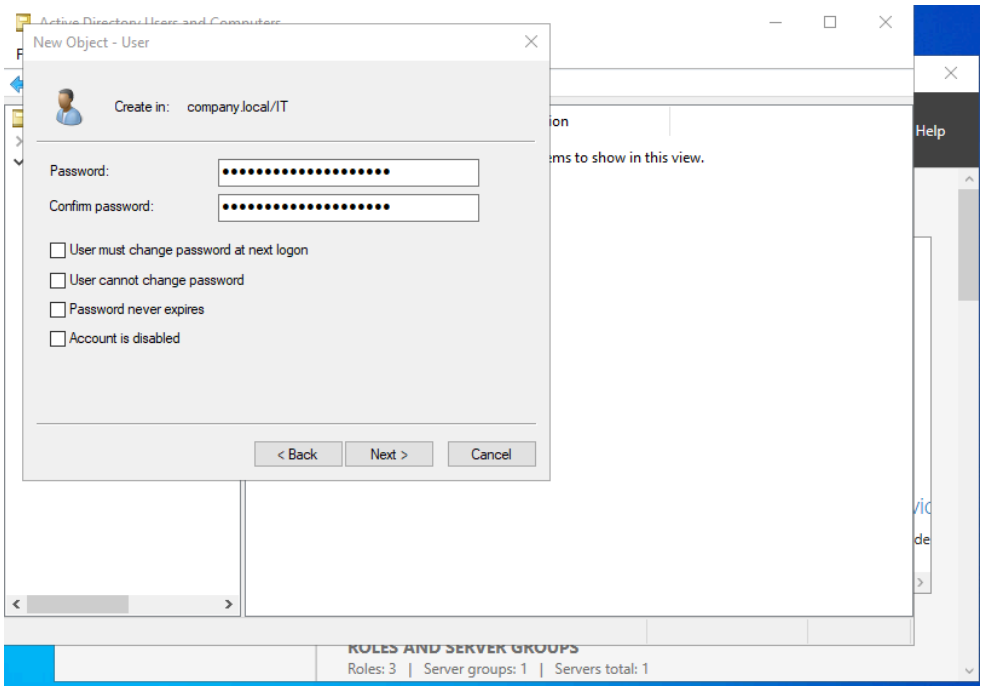
Right clicked and pressed *New* > *User* to add a user to IT



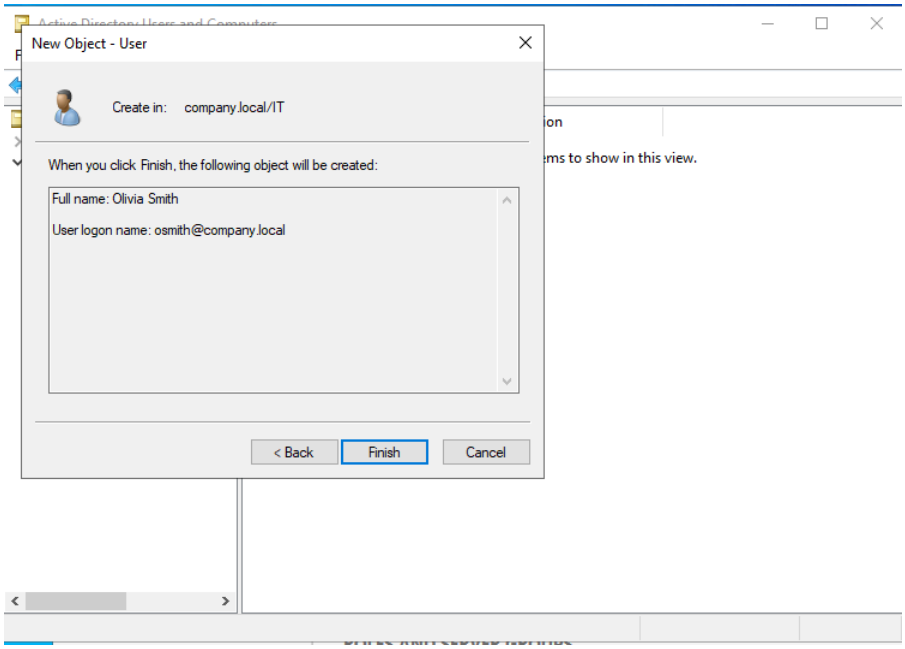
Filled out the necessary input fields to add the user and hit **Next**



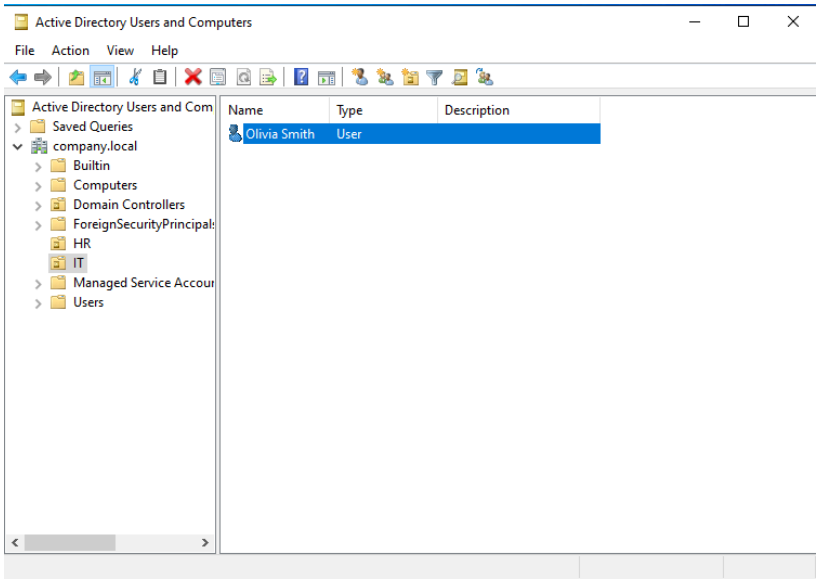
Created a password for user and unchecked *User must change password at next logon* and then I hit **Next**



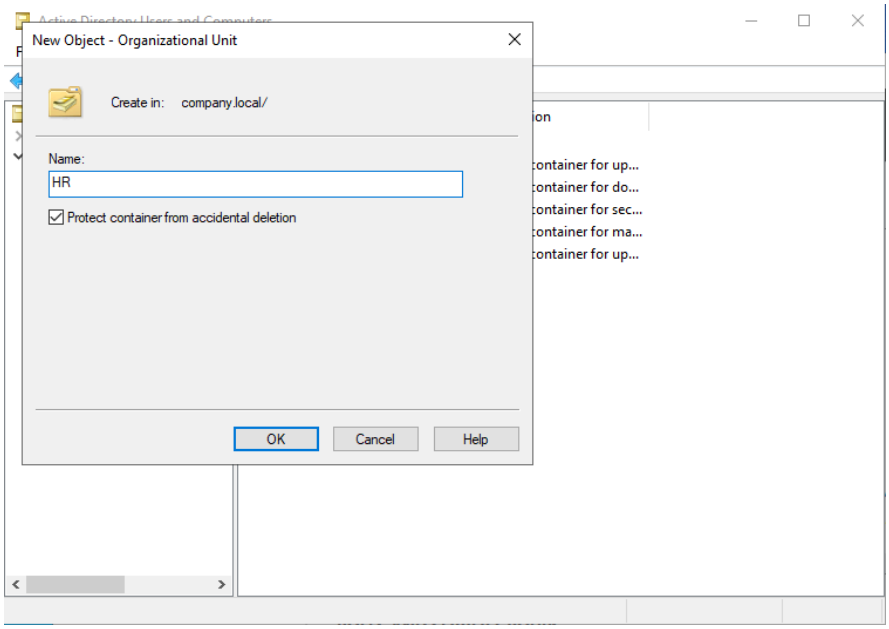
Hit **Finish**

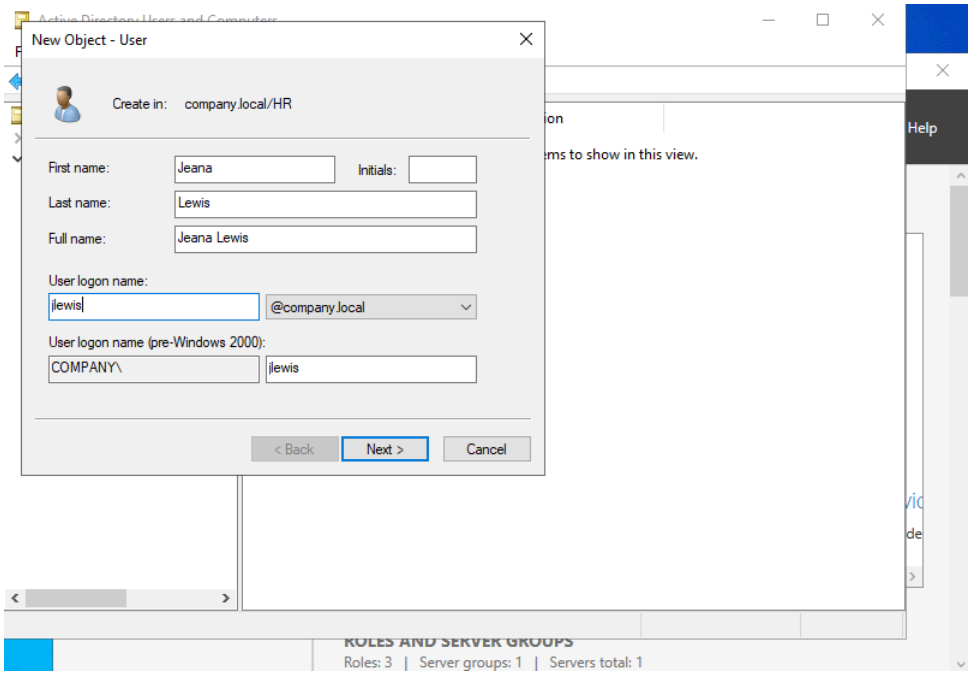


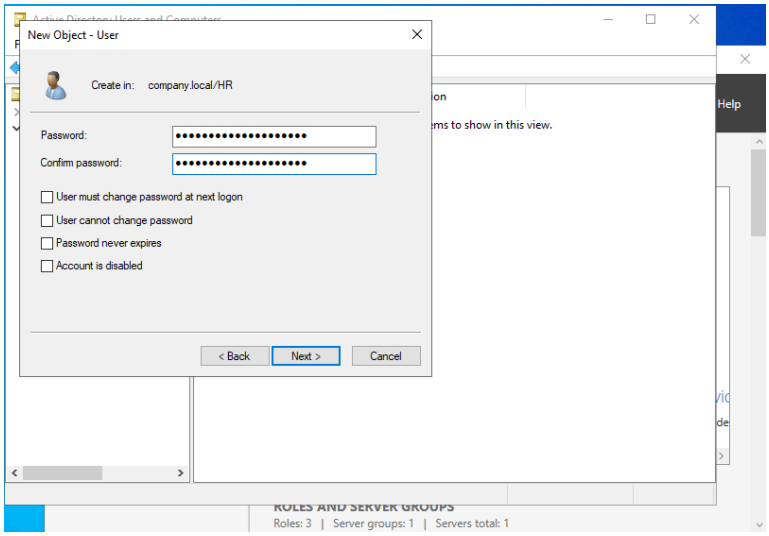
The user was added to the Organizational Unit IT successfully

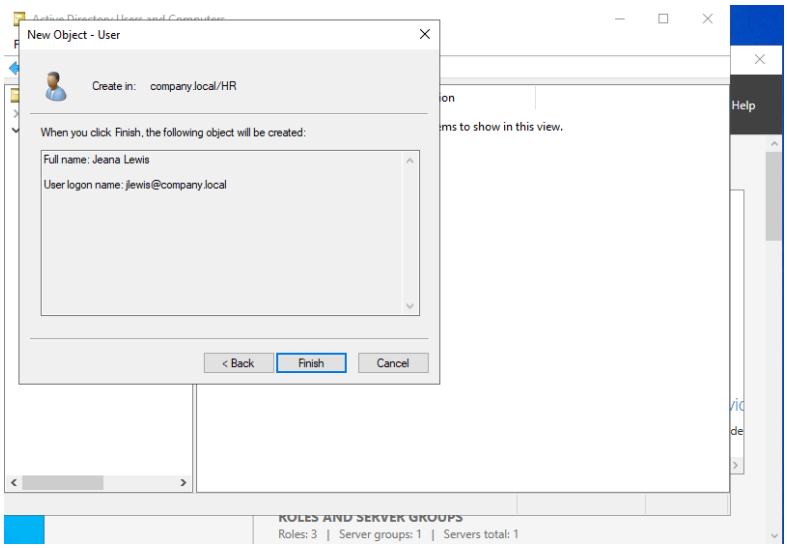


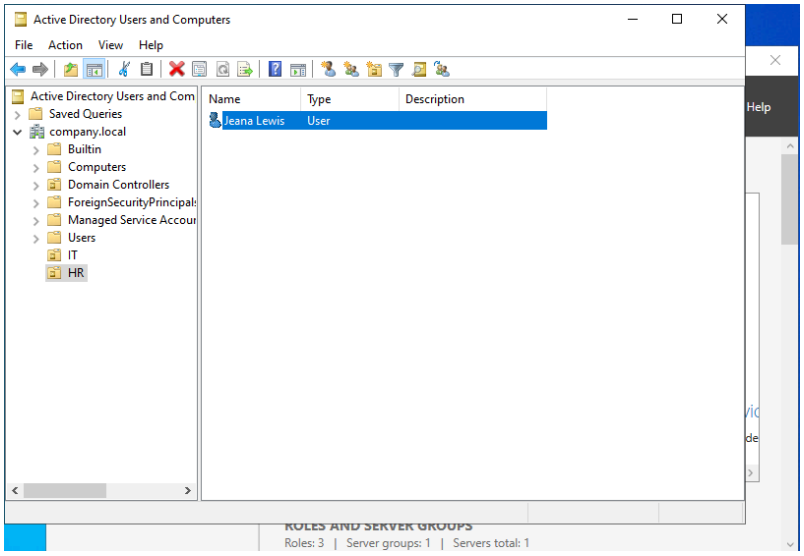
Added another Organization Unit HR and followed the same step as earlier to add user as I did for IT



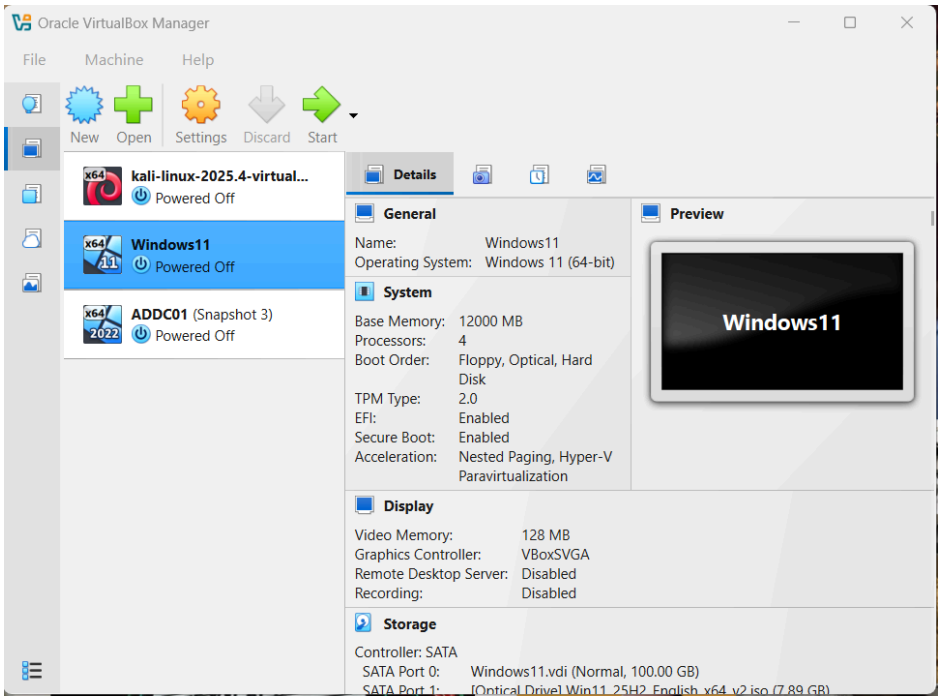

New Object - User
Create in: company.local/HR
First name: Jeana
Initials:
Last name: Lewis
Full name: Jeana Lewis
User logon name:
jlewis
@company.local
User logon name (pre-Windows 2000):
COMPANY\
jlewis
< Back
Next >
Cancel
Help
ROLES AND SERVER GROUPS
Roles: 3 | Server groups: 1 | Servers total: 1


New Object - User
Create in: company.local/HR
Password:
Confirm password:
User must change password at next logon
User cannot change password
Password never expires
Account is disabled
< Back
Next >
Cancel
Help
ms to show in this view.
ROLES AND SERVER GROUPS
Roles: 3 | Server groups: 1 | Servers total: 1

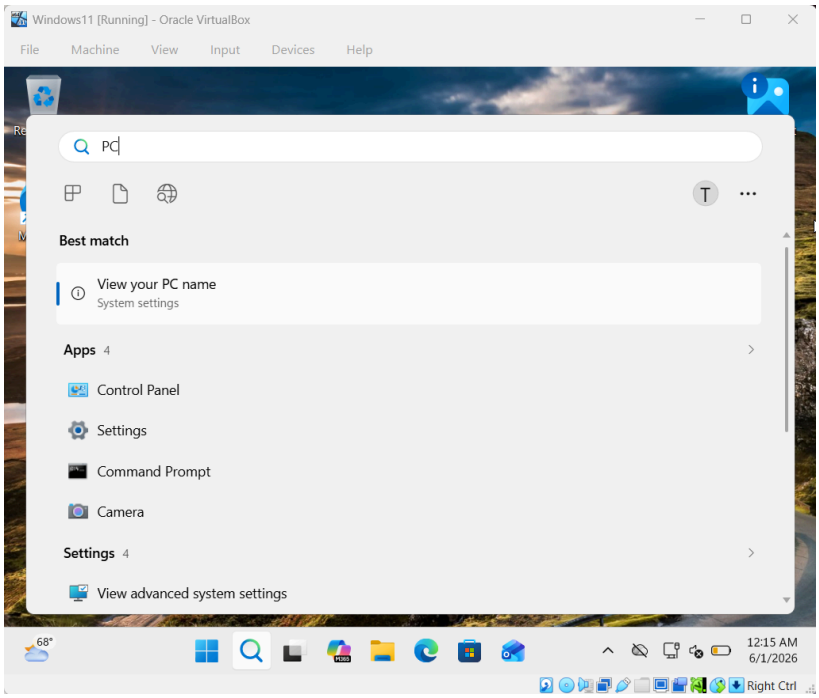

New Object - User
Create in: company.local/HR
When you click Finish, the following object will be created:
Full name: Jeana Lewis
User logon name: jlewis@company.local
< Back
Finish
Cancel
Help
ms to show in this view.
ROLES AND SERVER GROUPS
Roles: 3 | Server groups: 1 | Servers total: 1


Active Directory Users and Computers
File Action View Help
Active Directory Users and Com
Saved Queries
company.local
Builtin
Computers
Domain Controllers
ForeignSecurityPrincipals
Managed Service Accour
Users
IT
HR
Name
Type
Description
Jeana Lewis
User
Help
ROLES AND SERVER GROUPS
Roles: 3 | Server groups: 1 | Servers total: 1

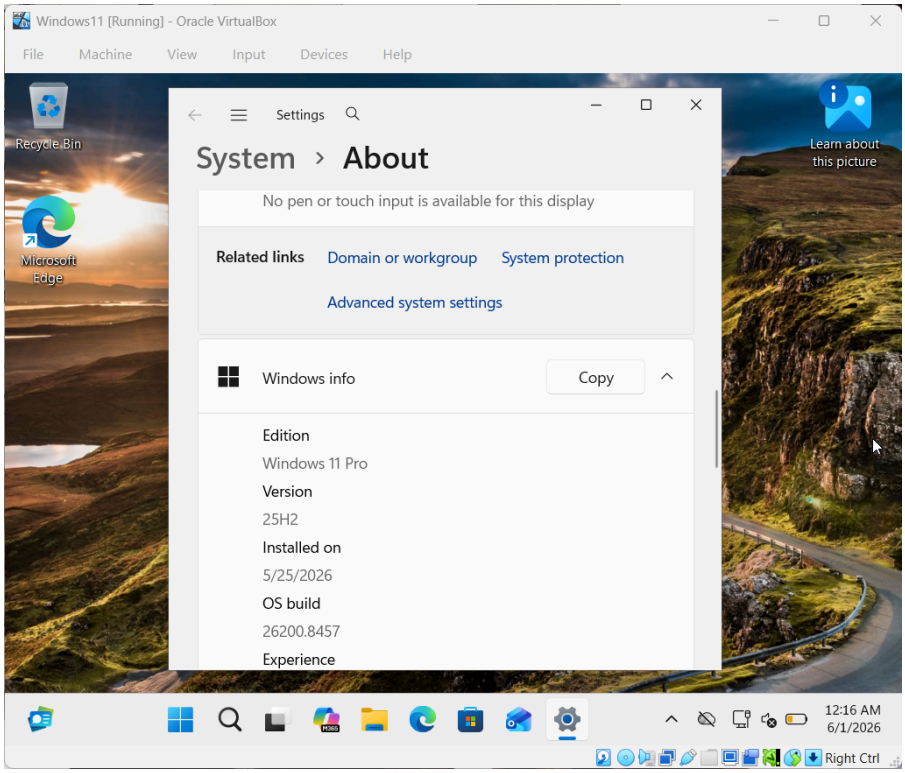
To log into one of these users account that I created. I went back to Windows 11 and started the machine up



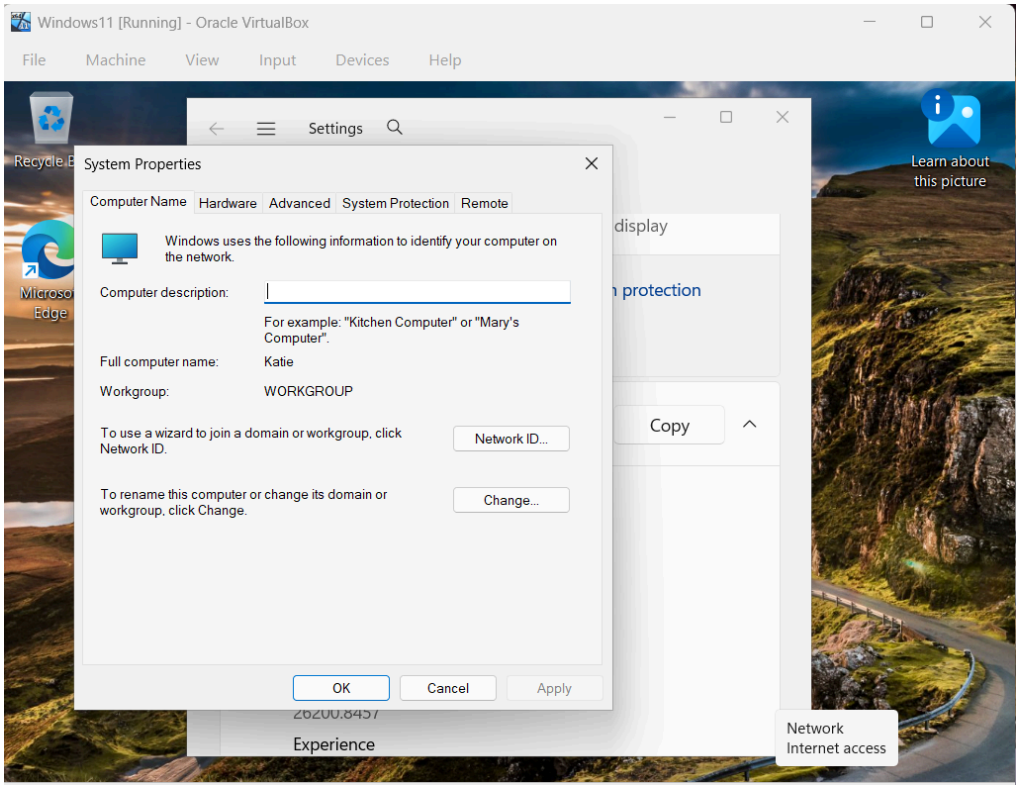
Once logged in, I typed in PC in the search bar and clicked on *View your PC name*



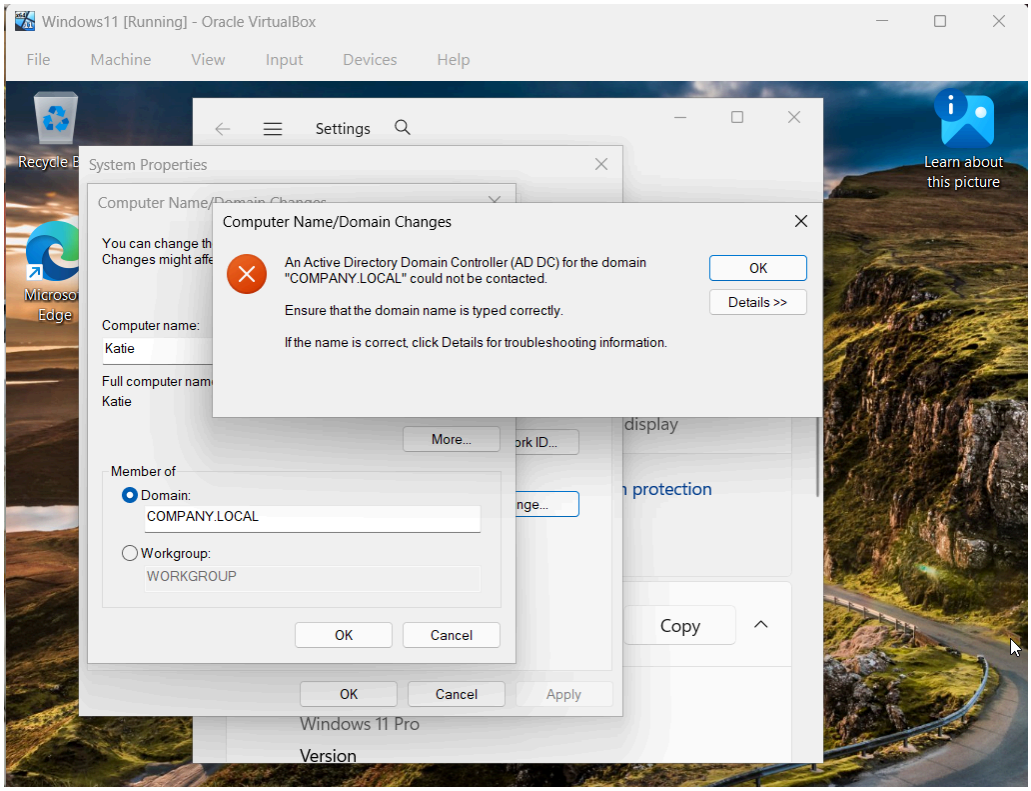
Scrolled down and clicked on *Advanced system settings*



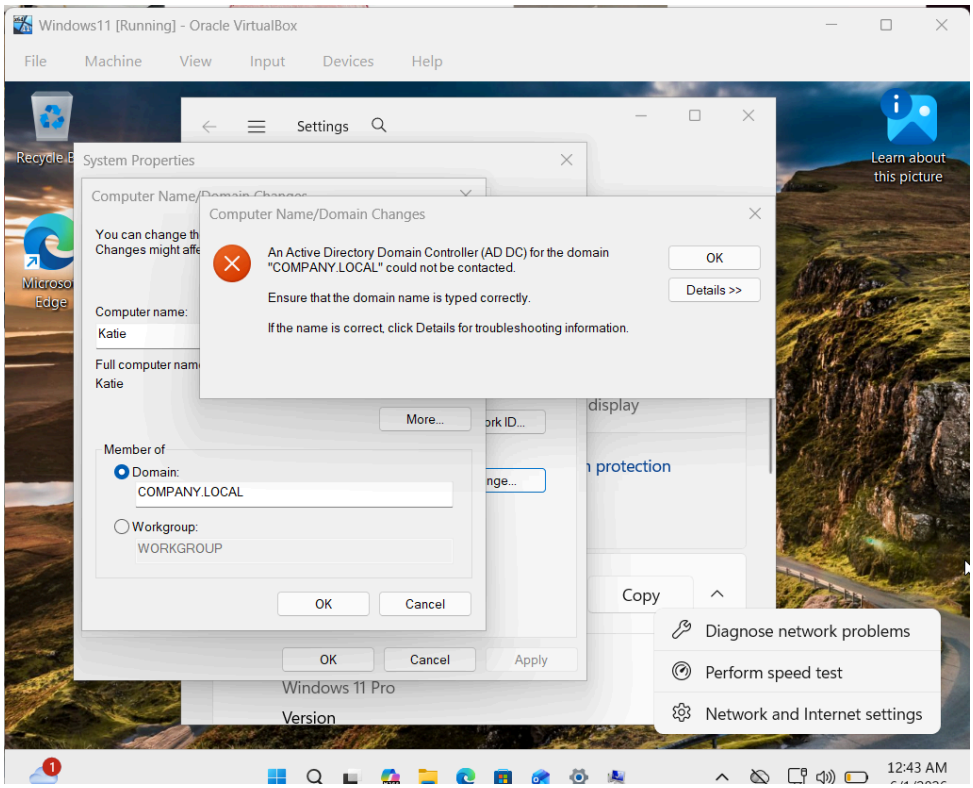
Clicked on *Computer Name > Change…*



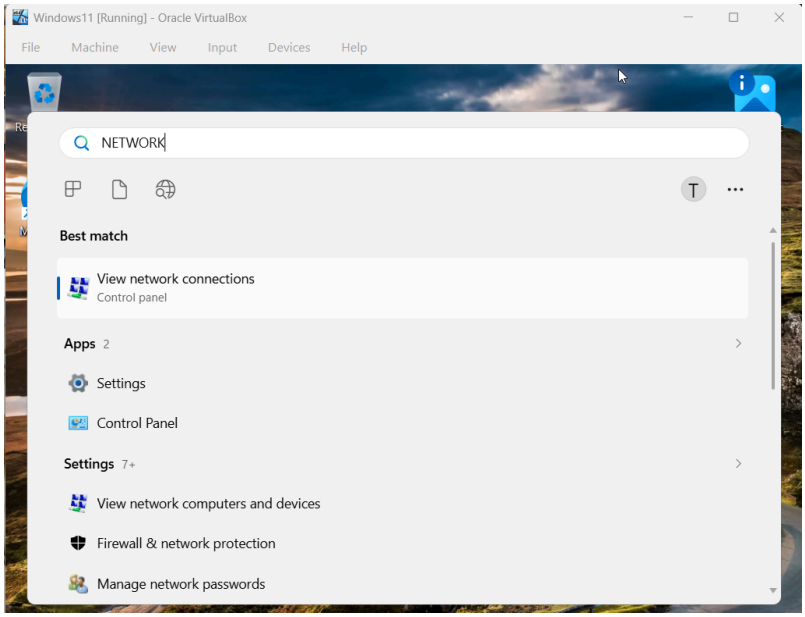
Click on the circle next to *Domain* so that I can input the domain name and hit **OK**



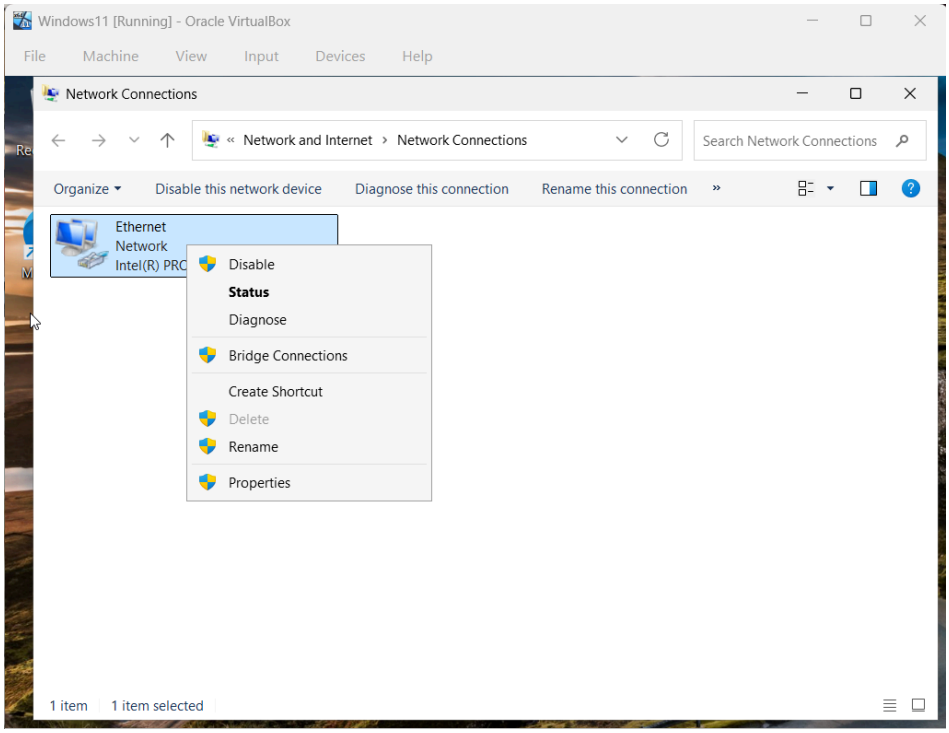
I came across this error message. To resolve this error, I went right clicked on the Network Icon



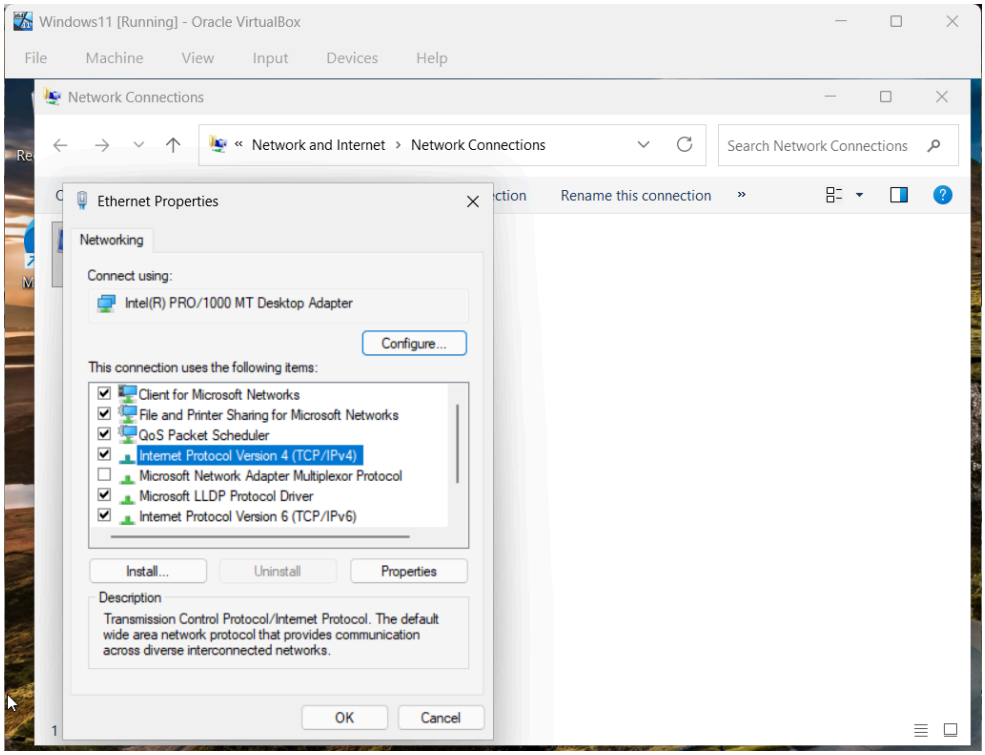
Typed Network in the search bar and clicked *View network connections*



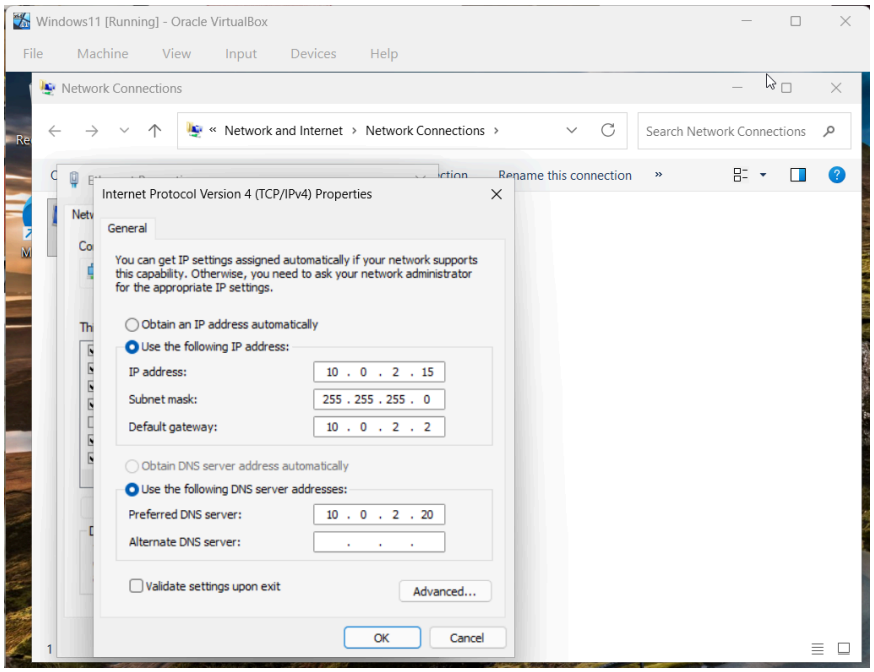
Right clicked Ethernet and pressed *Properties*



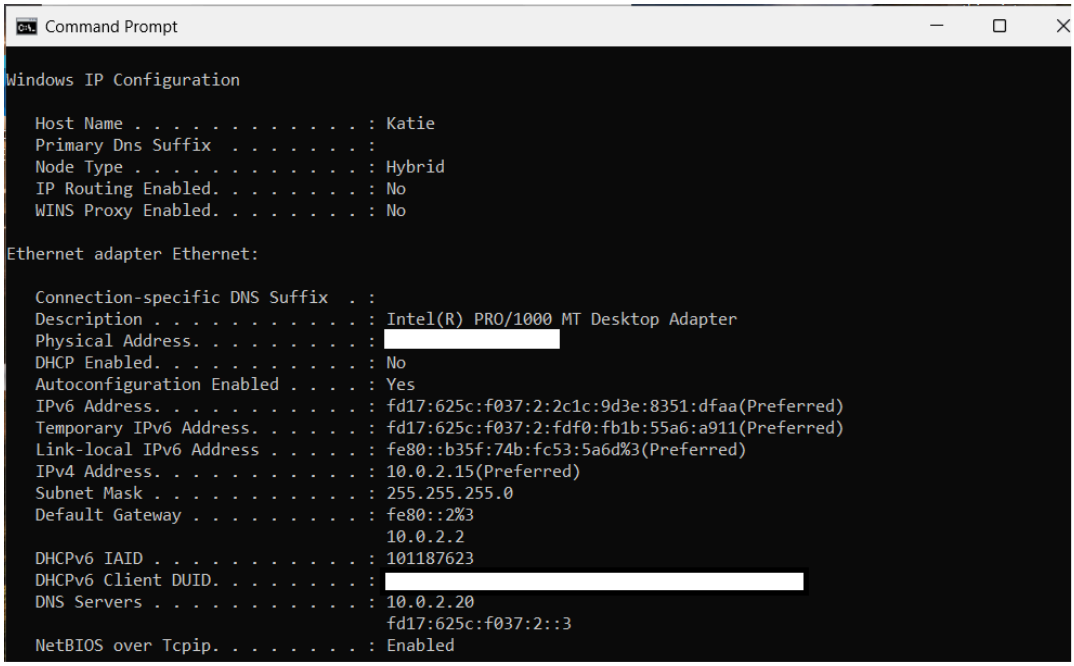
Clicked on *Internet Protocol Version 4* and hit **Properties**



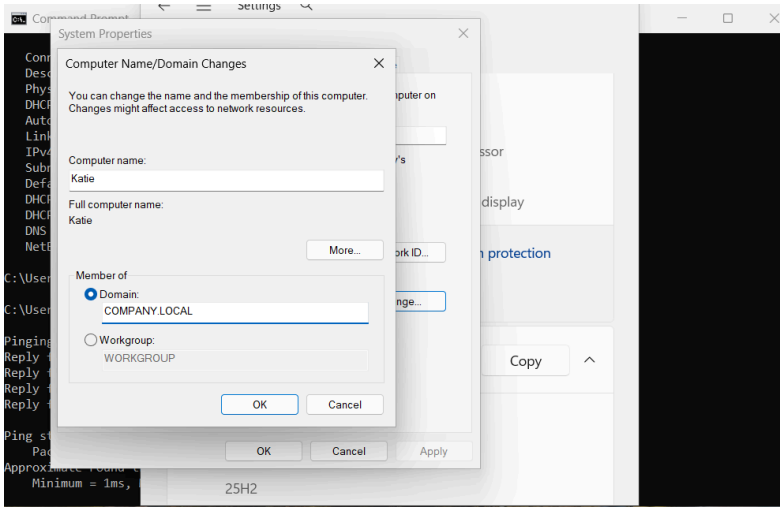
Filled out the necessary input fields and clicked **OK**

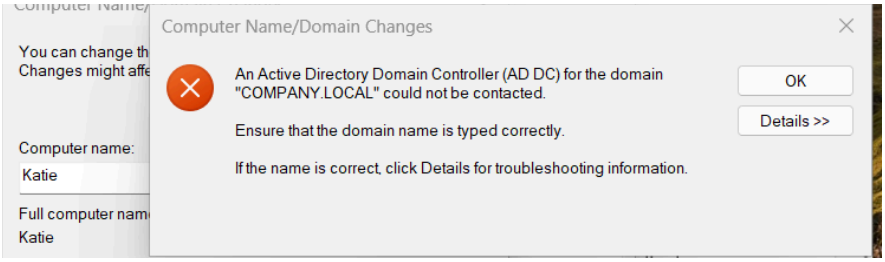


Checked that all the settings updated in the system by going to *command prompt* and typing *ipconfig/all.* It shows the DNS Server has been added

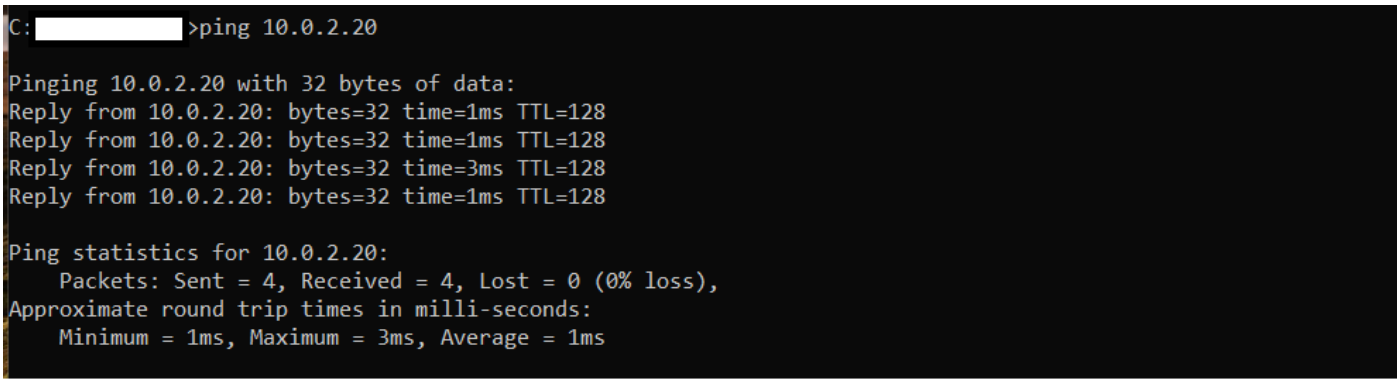


Then I went back to *System* > *About* > *Advanced system setting* retyped in the appropriate input and still got the same error message

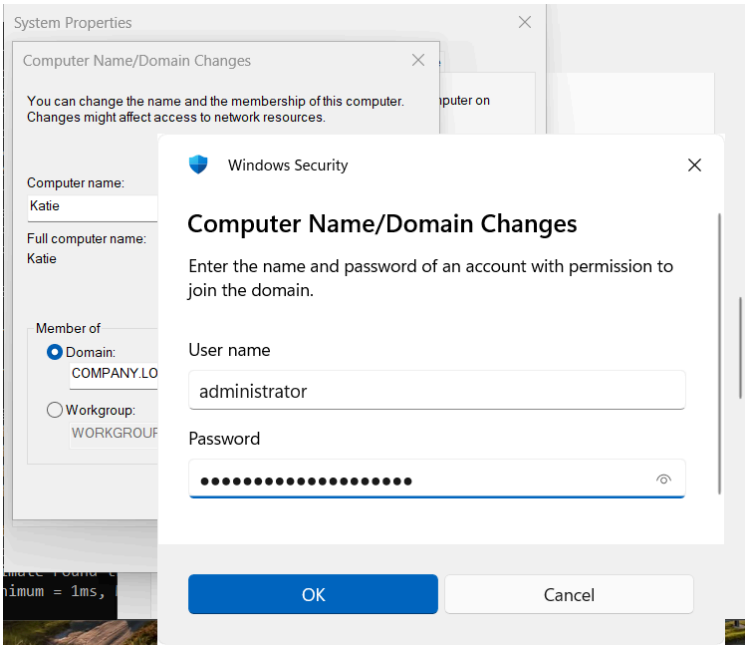




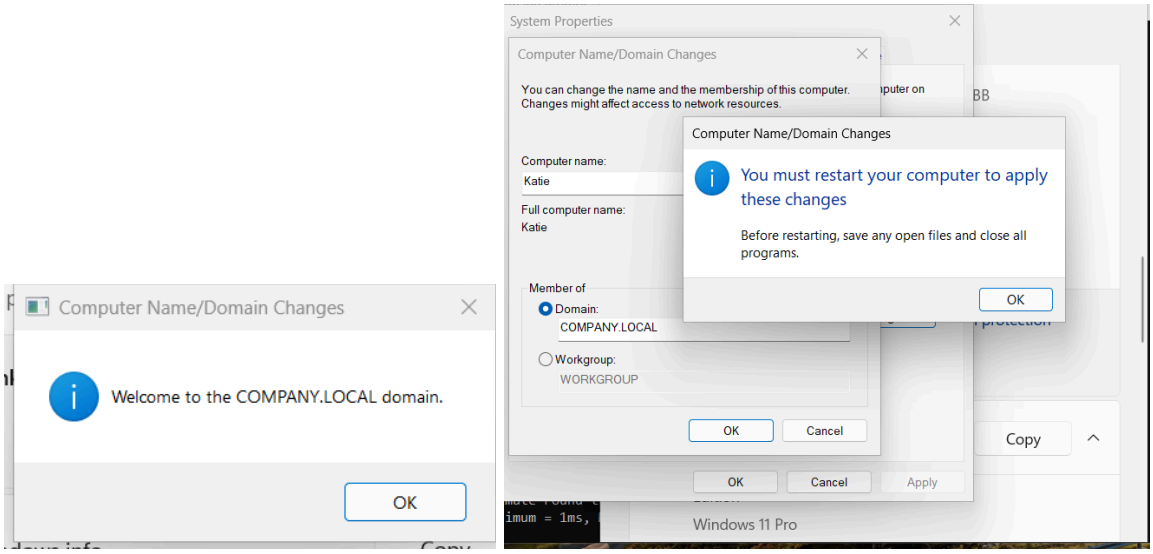
After further review, I realized the setting for NAT Network was not enabled on ADDC01 (Window Server 2022) nor the Windows 11 VM. Once I enabled setting for both, it allowed the VM's to interact with each other. I also pinged 10.0.2.20 from Windows 11 to verify connectivity.



Went back to *System* > *About* > *Advanced system setting* retyped in the appropriate input and still got the same error message one more time and was successful. Typed Administrator with the admin password I created earlier (ADDC01 password not DSRM)

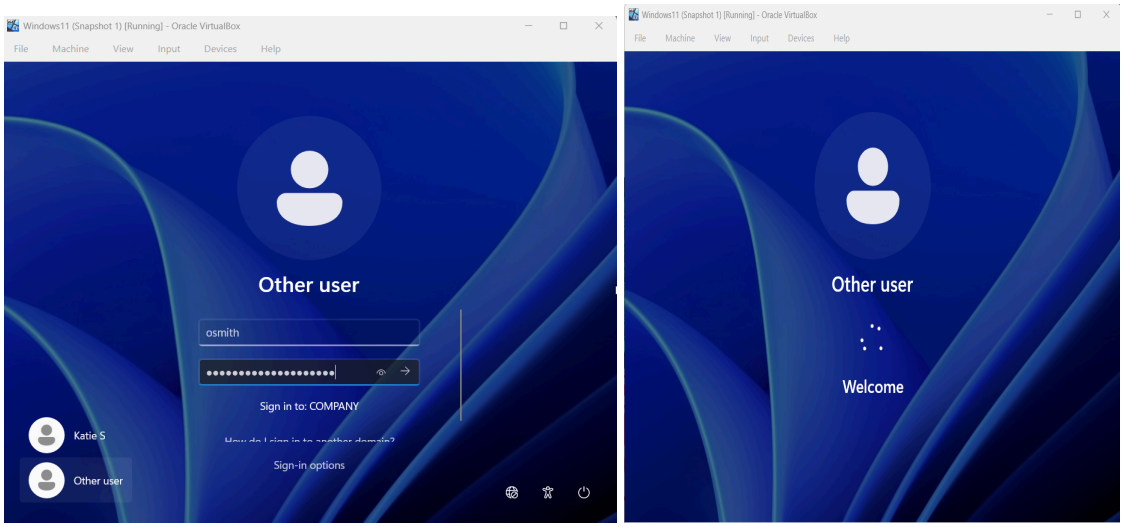


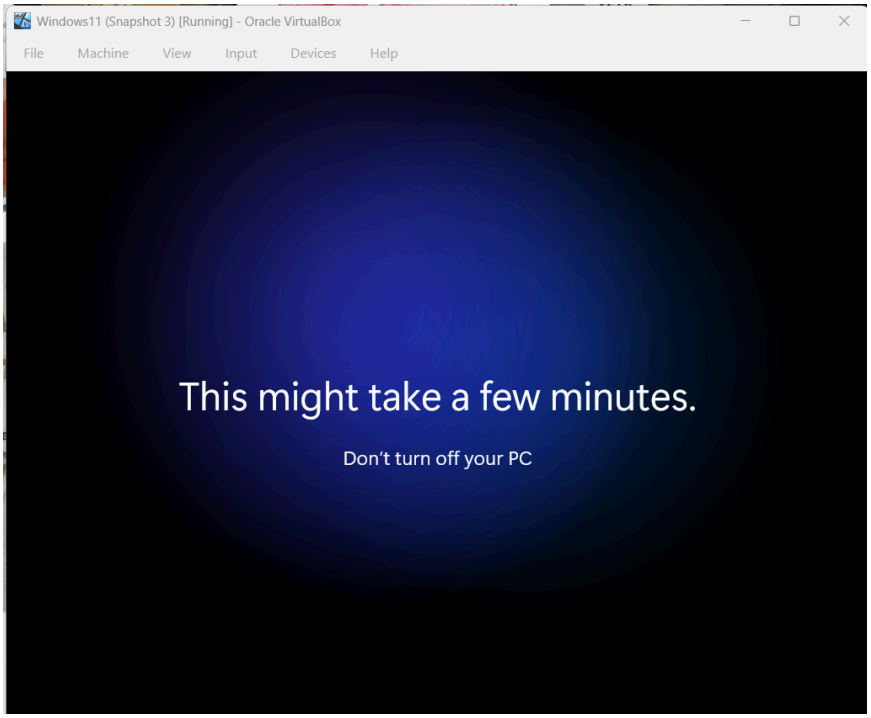
It successfully joined the domain. I clicked **OK** and closed all the files. Took a snapshot



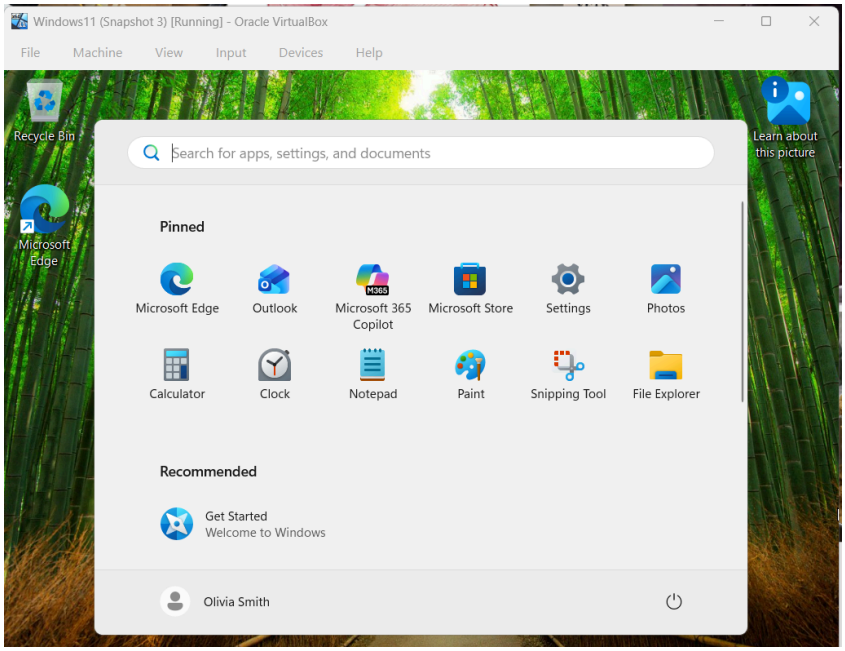
Then I hit **restart now** button

It restarted and brought me to the log in screen. I pressed *Other User* to try to sign on with another account.





I hit the Windows Icon, to confirm the user account I was in.



I successfully logged into Windows 11 using domain account osmith@company.local confirming Active Directory authentication is working correctly